

Each sub-contractor shall coordinate his work to avoid interference with work by other trades, and to insure that proper headroom and clearances are maintained as required by applicable codes.

All contractors to visit premises and become familiar with all of the work to be performed. No extra work orders will be allowed for failure to be informed. The architect shall be contacted if any discrepancies or conflicts are encountered.

② PROPOSED SITE PLAN
1/8" = 1'-0"

GRAPHIC SCALE



A horizontal scale bar with alternating black and white segments. The bar is divided into 8 equal segments, each 4 units long. The segments are labeled 0, 8, 16, 24, and 32 at the bottom.



GENERAL NOTES

1. ALL SUBCONTRACTORS SHALL VERIFY THAT THEY HAVE THE CURRENT PLANS PRIOR TO COMMENCEMENT OF ANY WORK.
2. VERIFY ALL DIMENSIONS AND SITE CONDITIONS IN THE FIELD PRIOR TO COMMENCEMENT OF ANY WORK.
3. ALL SUBCONTRACTORS ARE LIABLE FOR THE SAFETY OF THE EMPLOYEES AND EQUIPMENT WHILE ON THE JOB SITE.
4. ALL DIMENSIONS ARE FOR ROUGH FRAMING. WALLS ARE DIMENSIONED 3/16" UNLESS OTHERWISE NOTED.
5. ALL CONCRETE PIERS & FOOTINGS SHALL REST ON APPROVED BEARING 3000 PSF MIN. SOIL AND 3'-6" MIN. BELOW ADJOINING GRADE OR AS NOTED. IT SHALL BE THE CONTRACTORS RESPONSIBILITY TO VERIFY SOILS BEARING CAPACITY THROUGH RECORDED INFORMATION AND/OR SOILS EVALUATION
6. ALL FOUNDATION WALLS, FOOTING, AND PIERS SHALL BE POURED CONCRETE W/A MIX OF 1:2:5, 3000 PSI 28 DAY TEST.
7. PROVIDE DOUBLE FRAMING AT ALL OPENINGS; DOUBLE JOISTS UNDER ALL PARTITIONS PARALLEL TO JOISTS AS NOTED.
8. (2) 2 X 12 HEADER IS TYPICAL FOR DOOR/WINDOW OPENINGS UNLESS OTHERWISE NOTED.
9. GLUE AND NAIL ALL MICROLAM HEADERS.
10. ALL OPERABLE WINDOWS SHALL HAVE SCREENS.

FIREBLOCKING NOTES

1. FIRESTOP ALL DUCT, PIPE AND FLUE PENETRATIONS.
2. MAINTAIN AND MATCH FIRE RATING OF ASSEMBLY BEING PENETRATED.
3. FIREBLOCK FLOOR CAVITIES AT WALL AND OPENINGS IN FLOOR ASSEMBLIES.

LIFE SAFETY NOTES

1. ALL GLAZING MORE THAN 18" IN WIDTH AND ADJACENT TO ANY DOOR AND WITH SILL HEIGHT OF LESS THAN 24" ABOVE FINISH FLOOR SHALL BE TEMPERED SAFETY GLASS.
2. ALL INTERIOR FINISHES TO BE CLASS 'A' WITH A FLAME SPREAD RATING PER THE CHICAGO CONSTRUCTION CODE (SECTION 803.1.2).
3. ALL WOOD BLOCKING TO BE FIRED RATED.
4. ALL SMOKE DETECTORS AND CARBON MONOXIDE DETECTORS TO BE 120V HARDWIRED.

FOUNDATION NOTES

INSTALL A PASSIVE SUB-SLAB RADON CONTROL SYSTEM: ALIGNED UP WITH PLUMBING WALL UP ROOF. PROVIDE 4" PVC VENT PIPE FROM UNDER NEW SLAB (WITH A PIPE "T" FOR SUPPORT) AND UP TO ROOF EXTENDED 12" MIN. ABOVE ROOF AND NO LESS THAN 10' FROM ANY OPENINGS INTO CONDITIONED SPACES. SEAL ALL OPENING, GAPS, AND JOINTS IN FLOOR AND WALL ASSEMBLIES IN CONTACT WITH SOIL AND GAPS AROUND PIPES, DRAIN, ETC. PENETRATE THESE ASSEMBLIES.

PLAN LEGEND

EXISTING

NEW

SIC

SMOKE DETECTOR HARD WIRED INTERCONNECTED W/BATTERY BACK UP

CM

CARBON MONOXIDE HARD WIRED INTERCONNECTED W/BATTERY BACK UP

FD

FLOOR DRAIN

A

UL1305-1 HE INTERIOR PARTITION

NEW (2)2X4 PLATES

NEW 5/8" GYP BD TYPE "C", EA SIDE

NEW FIRE RETARDANT 2X4 WD STUD @ 16" O.C.

NEW 2X4 SILL PLATE

B

UL1305-1 HE INTERIOR PARTITION

NEW (2)2X6 PLATES TO REMAIN

NEW 5/8" GYP BD, EA SIDE

NEW FIRE RETARDANT 2X6 WD STUD @ 16" O.C. TO REMAIN

NEW 2X6 SILL PLATE TO REMAIN

FOUNDATION CORNER DETAIL

LAP 48"

CORNER BARS MATCH SIZE & SPA. OF HORIZ. BARS, WHERE SIZE AND SPA. DIFFER MATCH LARGER AREA. PROVIDE #4 @ 12" O.C. CORNER BARS WHERE THERE ARE NO HORIZ. BARS.

NOTES:
1. CONTRACTOR TO COORDINATE WITH WATER DEPARTMENT & GAS COMPANY FOR NEW GAS LINE INSTALLATION & LOCATION

1 PROPOSED FOUNDATION PLAN
1/4" = 1'-0"

PLAN LEGEND:

- SIC: SMOKE DETECTOR HARD WIRED INTERCONNECTED W/BATTERY BACK UP
- CM: CARBON MONOXIDE HARD WIRED INTERCONNECTED W/BATTERY BACK UP
- FD: FLOOR DRAIN

FOUNDATION CORNER DETAIL:

CORNER BARS MATCH SIZE & SPA. OF HORIZ. BARS, WHERE SIZE AND SPA. DIFFER MATCH LARGER AREA. PROVIDE #4 @ 12" O.C. CORNER BARS WHERE THERE ARE NO HORIZ. BARS.

2 PROPOSED 1ST FLOOR PLAN
1/4" = 1'-0"

ROOMS AND AREAS:

- GARAGE: 178 SF
- KITCHEN: 004, 178 SF
- DINING ROOM: 003, 160 SF
- LIVING ROOM: 001, 247 SF
- FAMILY ROOM: 002, 200 SF
- POWDER ROOM: 005, 30 SF
- STORAGE: 86
- PANTRY: 86
- LANDING: 3'-0"
- CONCRETE STOOP: 12A, 17

GRAPHIC SCALE: 0 8 16 24 32

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3. ALL SUBCONTRACTORS ARE LIABLE FOR THE SAFETY OF THE EMPLOYEES AND EQUIPMENT WHILE ON THE JOB SITE.

4. ALL DIMENSIONS ARE FOR ROUGH FRAMING. WALLS ARE DIMENSIONED 3 1/2" UNLESS OTHERWISE NOTED.

5. ALL CONCRETE PIERS & FOOTINGS SHALL REST ON APPROVED BEARING 3000 PSF MIN. SOIL AND 3'-6" MIN. BELOW ADJOINING GRADE OR AS NOTED. IT SHALL BE THE CONTRACTORS RESPONSIBILITY TO VERIFY SOILS BEARING CAPACITY THROUGH RECORDED INFORMATION AND/OR SOILS EVALUATION

6. ALL FOUNDATION WALLS, FOOTING, AND PIERS SHALL BE POURED CONCRETE W/A MIX OF 1:2:5, 3000 PSI 28 DAY TEST.

7. PROVIDE DOUBLE FRAMING AT ALL OPENINGS; DOUBLE JOISTS UNDER ALL PARTITIONS PARALLEL TO JOISTS AS NOTED.

8. (2) 2 X 12 HEADER IS TYPICAL FOR DOOR/WINDOW OPENINGS UNLESS OTHERWISE NOTED.

9. GLUE AND NAIL ALL MICROLAM HEADERS.

10. ALL OPERABLE WINDOWS SHALL HAVE SCREENS.

FIREBLOCKING NOTES

1. FIRESTOP ALL DUCT, PIPE AND FLUE PENETRATIONS.

2. MAINTAIN AND MATCH FIRE RATING OF ASSEMBLY BEING PENETRATED.

3. FIREBLOCK FLOOR CAVITIES AT WALL AND OPENINGS IN FLOOR ASSEMBLIES.

LIFE SAFETY NOTES

1. ALL GLAZING MORE THAN 1'8" IN WIDTH AND ADJACENT TO ANY DOOR AND WITH SILL HEIGHT OF LESS THAN 24" ABOVE FINISH FLOOR SHALL BE TEMPERED SAFETY GLASS.

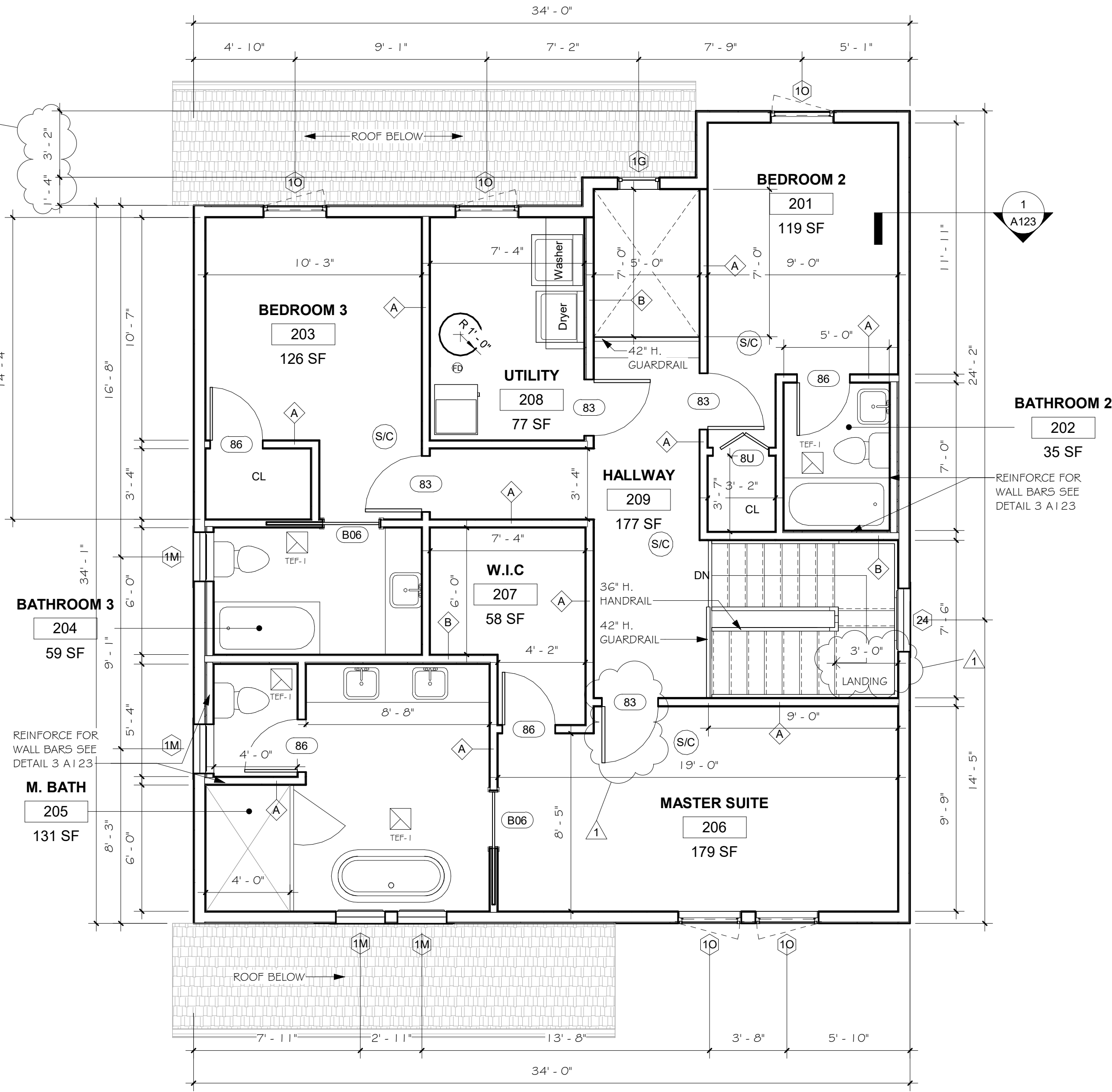
2. ALL INTERIOR FINISHES TO BE CLASS 'A' WITH A FLAME SPREAD RATING PER THE CHICAGO CONSTRUCTION CODE (SECTION 803.1.2).

3. ALL WOOD BLOCKING TO BE FIRED RATED.

4. ALL SMOKE DETECTORS AND CARBON MONOXIDE DETECTORS TO BE 120V HARDWIRED.

FOUNDATION NOTES

INSTALL A PASSIVE SUB-SLAB RADON CONTROL SYSTEM: ALIGNED UP WITH PLUMBING WALL UP ROOF. PROVIDE 4" PVC VENT PIPE FROM UNDER NEW SLAB (WITH A PIPE "T" FOR SUPPORT) AND UP TO ROOF EXTENDED 12" MIN. ABOVE ROOF AND NO LESS THAN 1'0" FROM ANY OPENINGS INTO CONDITIONED SPACES. SEAL ALL OPENING, GAPS, AND JOINTS IN FLOOR AND WALL ASSEMBLIES IN CONTACT WITH SOIL AND GAPS AROUND PIPES, DRAIN, ETC. PENETRATE THESE ASSEMBLIES.



1 PROPOSED 2ND FLOOR PLAN
1/4" = 1'-0"

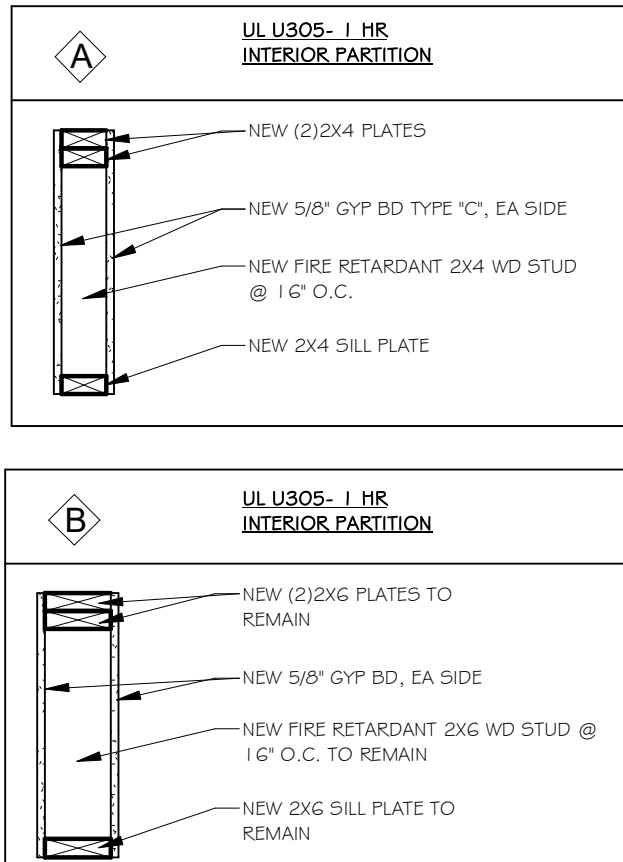
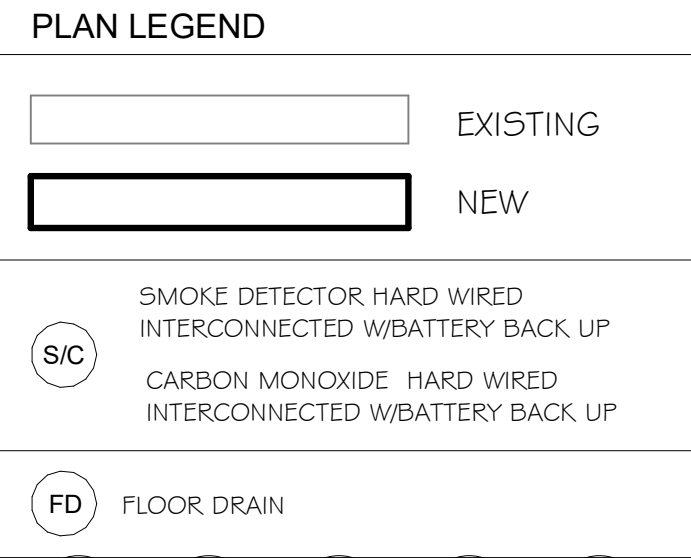


TABLE 5-52

FACTORED UNIFORM LOAD - MAXIMUM FACTORED UNIFORM LOAD APPLIED TO EITHER OUTSIDE MEMBER (PLF)

April 2024 (Expires 4/2026)

Fastener Type	Placement	Number of Rows	Fastener On-Center Spacing	Fastener Pattern		
				Assembly A	Assembly B	Assembly F
10d (0.128" x 3") or (0.131" x 3") Nail ^[1]	As shown	3	12"	865	650	
		4	12"	1,150	880	780
		3	19.2"	1,465	1,095	975
		4	19.2"	1,950	1,465	1,300
1/2" A307 Through Bolt ^[2]	As shown	3	12"	1,755	1,315	1,170
		4	12"	2,340	1,755	1,560
		3	19.2"	2,175	1,630	1,700
		4	19.2"	2,610	1,960	2,040
Simpson Strong-Tie® SDS ^[3]	As shown	3	12"	1,305	980	1,020
		4	12"	1,630	1,225	1,275
		3	19.2"	1,950	1,470	1,530
		4	19.2"	2,175	1,630	1,700
MiTek® WS ^[4]	As shown	3	12"	1,355	1,015	1,150
		4	12"	1,695	1,270	1,440
		3	19.2"	2,035	1,525	1,725
		4	19.2"	2,260	1,695	1,915
Simpson Strong-Tie® SDW22 ^[5]	One face	3	12"	1,020	935	830
		4	12"	1,275	1,165	1,040
		3	19.2"	1,360	1,245	1,105
		4	19.2"	1,700	1,555	1,385

[1] Nailed connection values may be doubled for 6" on-center or tripled for 4" on-center nail spacing.

[2] Washer required. Bolt holes to be 3/16" maximum.

[3] Factored resistance for 24" on-center bolted or screwed connection values may be doubled for 12" on-center spacing.

[4] When loading the head side of a SDW22 screw, assemblies B and F can be increased by 15%.

[5] Bold italic loads indicate assemblies that require fastener placement on both faces. Stagger fasteners on the second face so they fall halfway between fasteners on the first face.

Simpson Strong-Tie® Wood Construction Connectors

CC/ECC/ECU

Column Caps (cont.)

These products are available with additional corrosion protection. For more information, see pp. 13-19. For stainless steel fasteners, see p. 25.

Model No.	Beam Width (in.)	W ₁ (in.)	W ₂ (in.)	Dimensions (in.)		Bolts		DFWP Allowable Loads				Code Ref.	CC/ECC/ECU Model No. (No. Legs)				
				L	H	CC	ECG	ECU	ECU	ECU	ECU						
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	16,300	6,875	5,150	6,875	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	21,485	10,760	5,150	10,760	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
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CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
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CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
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CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4	2	4	2	5,150	24,080	12,040	5,150	12,040	CC104-1A, ECC104-1A
CC104	3 1/2	3 1/2	3 1/2	11	7 1/4	9 1/4	9 1/4	4									



1 PROPOSED FRONT ELEVATION
1/4" = 1'-0"



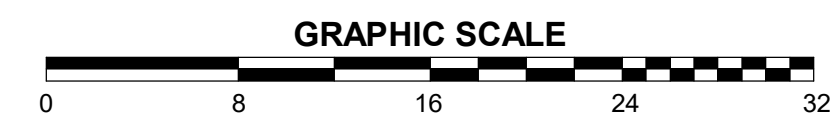
2 PROPOSED RIGHT ELEVATION
1/4" = 1'-0"



3 PROPOSED REAR ELEVATION
1/4" = 1'-0"



4 PROPOSED LEFT ELEVATION
1/4" = 1'-0"



ARCHITECTONIC
SOLUTIONS, INC.

4348 W LAKE ST STE 306
CHICAGO, IL 60624
312-373-0098

DESIGN FIRM NO. 184.006860-0001

I HEREBY CERTIFY THAT THESE
PLANS WERE PREPARED BY ME OR
UNDER MY DIRECT SUPERVISION AND
TO THE BEST OF MY KNOWLEDGE
AND BELIEF, THEY COMPLY WITH ALL
OF ROBBINS, ILLINOIS BUILDING
CODES AND ORDINANCES



SIGNATURE: *Oscar Alba*

EXPIRES: 11/30/26

DATE	ISSUE
11/21/25	FOR REVIEW
1/12/26	FIELD REVISION

JOB TITLE

NEW
2 STORY FRAME
RESIDENCE

PROJECT ADDRESS

352 I 134th Pl.
Robbins, IL 60472

DRAWING TITLE

PROPOSED
ELEVATIONS

PROJECT # 352 I

ISSUE DATE: 21/11/25

DRAWN BY: P.M.

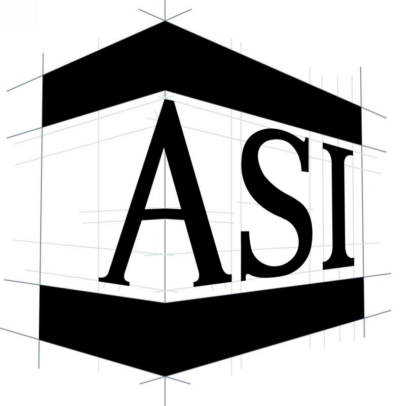
CHECKED BY: O. ALBA

SHEET

A122

2/12/2026 12:40:07 PM

THIS DRAWING IS DIAGRAMMATIC AND REPRESENTS
TYPICAL LOCATIONS OF REQUIRED FIREBLOCKING

ARCHITECTONIC
SOLUTIONS, INC.

4348 W LAKE ST STE 306
CHICAGO, IL 60624
312-373-0098

DESIGN FIRM NO. 184.006860-0001

I HEREBY CERTIFY THAT THESE
PLANS WERE PREPARED BY ME OR
UNDER MY DIRECT SUPERVISION AND
TO THE BEST OF MY KNOWLEDGE
AND BELIEF, THEY COMPLY WITH ALL
OF ROBBINS, ILLINOIS BUILDING
CODES AND ORDINANCES



SIGNATURE: Q. Ali

EXPIRES: 11/30/26

	DATE	ISSUE
	11/21/25	FOR REVIEW
①	1/12/26	FIELD REVISION
②		
③		
④		
⑤		

JOB TITLE

NEW
2 STORY FRAME
RESIDENCE

PROJECT ADDRESS

3521 134th Pl.
Robbins. IL 60472

DRAWING TITLE

FIRE BLOCKING

PROJECT #	3521
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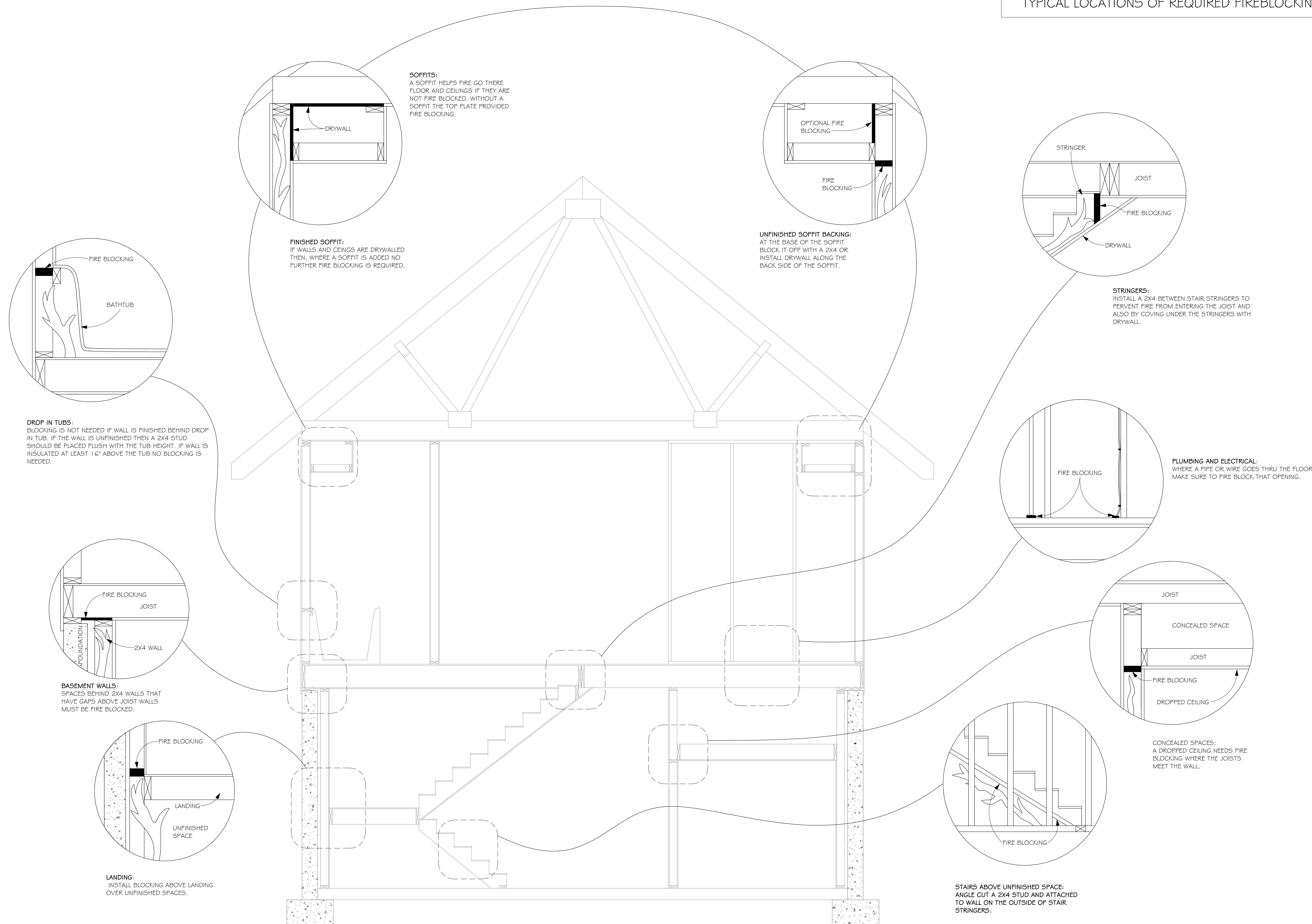
ISSUE DATE:	21/11/25
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DRAWN BY: P.M.

CHECKED BY:	O. ALBA
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SHEET

AI 70



1 FIRE BLOCKING
NOT TO SCALE

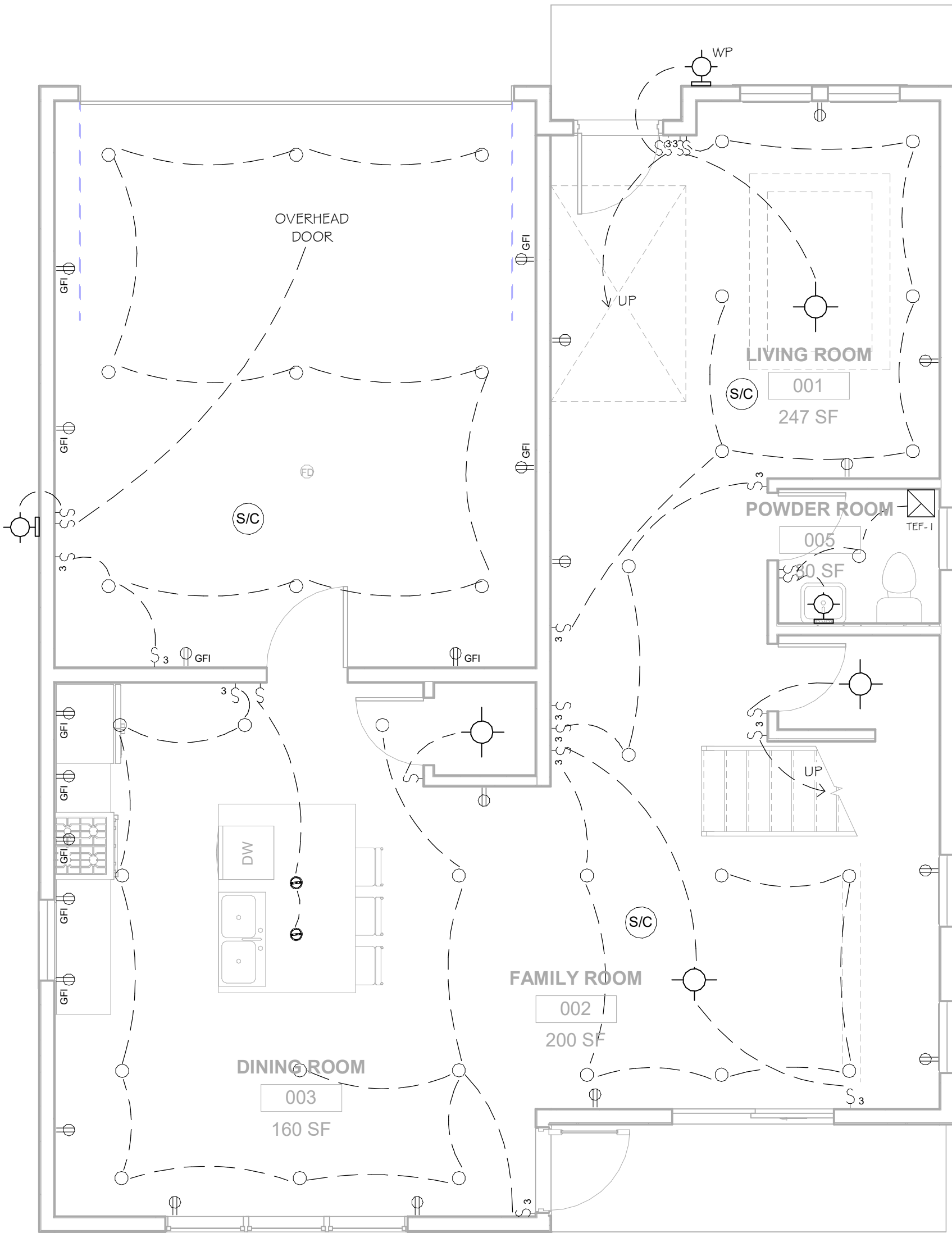
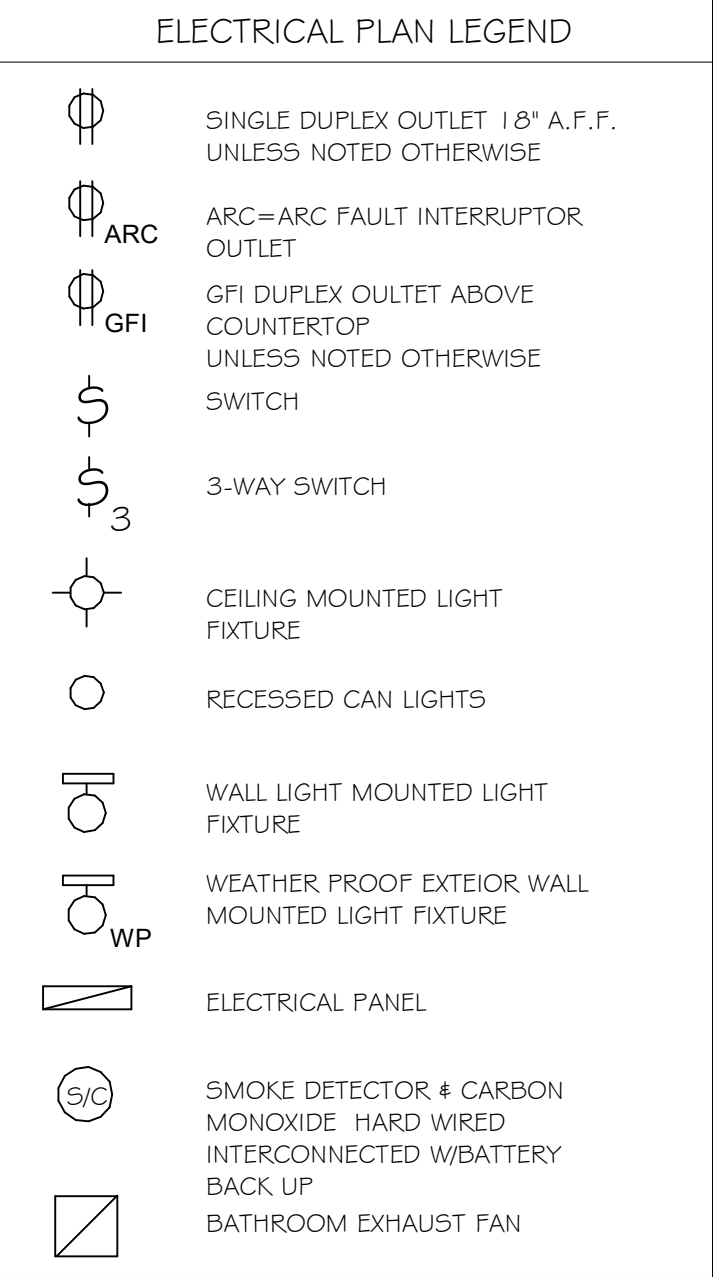
ELECTRICAL NOTES

1. THIS CONTRACTOR SHALL VISIT THE SITE AND BE FAMILIAR WITH THE JOB CONDITION PRIOR TO SUBMITTING THE BID.
2. THIS CONTRACTOR SHALL PAY FOR ALL THE TAXES, PERMITS AND FEES GIVE LOCAL AUTHORITIES ALL NOTICES RELATING TO HIS WORK.
3. THE ELECTRICAL CONTRACTOR SHALL INSTALL ALL WORK IN ACCORDANCE WITH ALL APPLICABLE STATE, LOCAL, AND NATIONAL CODES AND ORDINANCES.
4. THE ELECTRICAL CONTRACTOR SHALL PROVIDE COPPER WIRE THROUGHOUT. WIRING SHALL BE IN RIGID METAL CONDUIT OR EMT.
5. THIS CONTRACTOR SHALL GUARANTEE ALL WORK AND MATERIAL FOR ONE YEAR AFTER COMPLETION AGAINST ALL DEFECTS OF MATERIAL, EQUIPMENTS AND WORKMANSHIP.
6. DISTRIBUTION PANELS TO BE WITH 20 POLES SPACES MINIMUM PROVIDE BREAKERS AS REQUIRED.
7. BRANCH CIRCUITS:
A. FURNISH AND INSTALL BRANCH CIRCUITS FROM THE PANEL BOARD TO ALL THE CONVENIENCE OUTLETS AND LIGHTING FIXTURES AS SHOWN, MINIMUM. WIRE SIZE: #12
B. SERVE ALL APPLIANCES WITH INDIVIDUAL CIRCUITS #12 WIRE WERE REQUIRED.
8. WIRING DEVICES:
A. PLANS INDICATE APPROXIMATE OUTLET LOCATIONS. ALL OUTLETS SHALL BE GROUNDED DUPLEX PLUGS.
B. SWITCHES SHALL BE SINGLE POLE, 3 WAY, ETC., AS INDICATED.
C. ALL PLUGS AND SWITCHES SHALL BE IVORY UNLESS SPECIFIED OTHERWISE ALL COVER PLATES SHALL BE AS SELECTED BY THE OWNER.
9. PROVIDE AND INSTALL GROUND FAULT PROTECTED OUTLETS AT ALL BATHROOMS, KITCHENS, SHOWERS AND EXTERIOR OUTLETS.
10. PROVIDE AND INSTALL DISCONNECTING MEANS AS REQUIRED BY CODE FOR FIXED APPLIANCES.
11. SMOKE DETECTORS IN EVERY BEDROOM AND TOP OF ALL INTERIOR STAIRS. CARBON MONOXIDE DETECTORS AT ALL HEATING UNITS AND/OR ONE PER FLOOR AND BE UL APPROVED WITH AUDIBLE ALARM.
12. ALL CLOSET LIGHTS SHALL BE MOUNTED ON THE INSIDE WALL ABOVE THE CLOSET DOOR AND BE MINIMUM OF 18" AWAY FROM ALL SHELVING.
13. EXISTING AMP SERVICE TO REMAIN IF STATED IN PLANS.
14. PROVIDE TAMPER RESISTANT RECEPTACLES WHERE REQUIRED.
15. ELECTRICAL CONTRACTOR SHALL REMOVE ALL EXISTING RECEPTACLE, CONDUIT, LIGHTING FIXTURES AND DEVICES IN AREAS WHERE CEILING, WALLS, FLOORS OR PARTITION ARE RE-WORKED OR REMOVED. THE CONTRACTOR SHALL TAKE ADEQUATE CARE AND CAUTION AS NOT TO DAMAGE ADJACENT SURFACES.
16. ALL FIXTURES SHALL BE INSTALLED AT THE GEIGHT NOTED ON THE PLANS
17. THE ELECTRICAL CONTRACTOR TO PROVIDE SERVICE FOR USE DURING THE CONSTRUCTION PHASE.
18. THE ELECTRICAL CONTRACTOR TO TEST SYSTEM TO ENSURE ELECTRICAL SYSTEM.

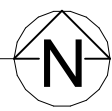
ELECTRICAL SPECIFICATIONS

1. THE GENERAL CONDITIONS AND SUPPLEMENTARY GENERAL CONDITIONS OF THE CONTRACT SHALL BE CONSIDERED TO BE INCLUDED AS PART OF THESE SPECIFICATIONS. THE ELECTRICAL CONTRACTOR SHALL CONSULT THEM IN DETAIL FOR INSTRUCTIONS PERTAINING TO THE WORK UNDER THIS TRADE.
2. THE ELECTRICAL CONTRACTOR SHALL FURNISH ALL LABOR AND MATERIAL NECESSARY FOR THE COMPLETE INSTALLATION OF ELECTRICAL WORK SHOWN ON THE PLANS AND SPECIFIED HEREINAFTER. THE INSTALLATION SHALL BE IN ACCORDANCE WITH ALL LOCAL CODES AND ALL APPLICABLE REQUIREMENTS OF THE UTILITY COMPANIES WHICH SERVE THE PROPERTY.
3. THE ELECTRICAL CONTRACTOR SHALL FURNISH, ERECT, INSTALL, CONNECT, CLEAN, ADJUST, TEST AND PLACE IN SERVICE ALL MATERIALS, EQUIPMENT AND SYSTEMS IN ACCORDANCE WITH THE MANUFACTURERS DIRECTIONS AND RECOMMENDATIONS FOR ALL WORK SHOWN ON THE DRAWINGS AND CALLED FOR IN THESE SPECIFICATIONS. APPARATUS, LIGHTING FIXTURES, MATERIALS OR WORK NOT SHOWN ON THE DRAWINGS BUT MENTIONED IN THESE SPECIFICATIONS, INCLUDING ANY INCIDENTAL ACCESSORIES NOT SPECIFIED OR SHOWN ON THE DRAWINGS BUT MENTIONED IN THESE SPECIFICATIONS, INCLUDING ANY INCIDENTAL ACCESSORIES NOT SPECIFIED OR SHOWN ON THE DRAWINGS NECESSARY TO MAKE THE WORK COMPLETE AN READY FOR OPERATION, SHALL BE FURNISHED AND INSTALLED WITHOUT ADDITIONAL EXPENSE TO THE OWNER. IT IS THE CONTRACTORS RESPONSIBILITY TO REVIEW ALL PROJECT DOCUMENTS, INCLUDING ARCHITECTURAL STRUCTURAL, MECHANICAL, PLUMBING, FIRE PROTECTION AND CIVIL DISCIPLINES.
4. THE ELECTRICAL CONTRACTOR SHALL PAY FOR ALL PERMIT FEES, PLAN, AND REVIEW FEES, LICENSE FEES AND INSPECTION FEES REQUIRED FOR THE EXECUTION OF HIS WORK. THE ELECTRICAL CONTRACTOR SHALL OBTAIN APPROVALS, WHERE REQUIRED, FROM ALL LOCAL INSPECTION AUTHORITIES FOR EXIT SIGNS, EMERGENCY LIGHTING AND FIRE ALARM SYSTEMS. HE SHALL PROVIDE AND SUBMIT TO THE INSPECTION AUTHORITIES ALL REQUIRED SUBMITTAL INFORMATION INCLUDING, BUT NOT LIMITED TO ALL PRELIMINARY DRAWINGS, DIAGRAMS, ZONE SCHEDULES, RISER DIAGRAMS, EQUIPMENT CATALOG SHEETS AND SYSTEM SPECIFICATIONS. HE SHALL ALSO BE RESPONSIBLE FOR THE FINAL SUBMISSION OF ALL DOCUMENTATION REQUIRED FOR THE FINAL APPROVAL OF OCCUPANCY INCLUDING, BUT NOT LIMITED TO POINT FIRE ALARM DRAWINGS AND CONDUIT SIZE AND WIRE COUNT ALL IN COMPLIANCE WITH THE SUBMITTAL REQUIREMENTS OF THE LOCAL ENFORCKING AGENCIES.

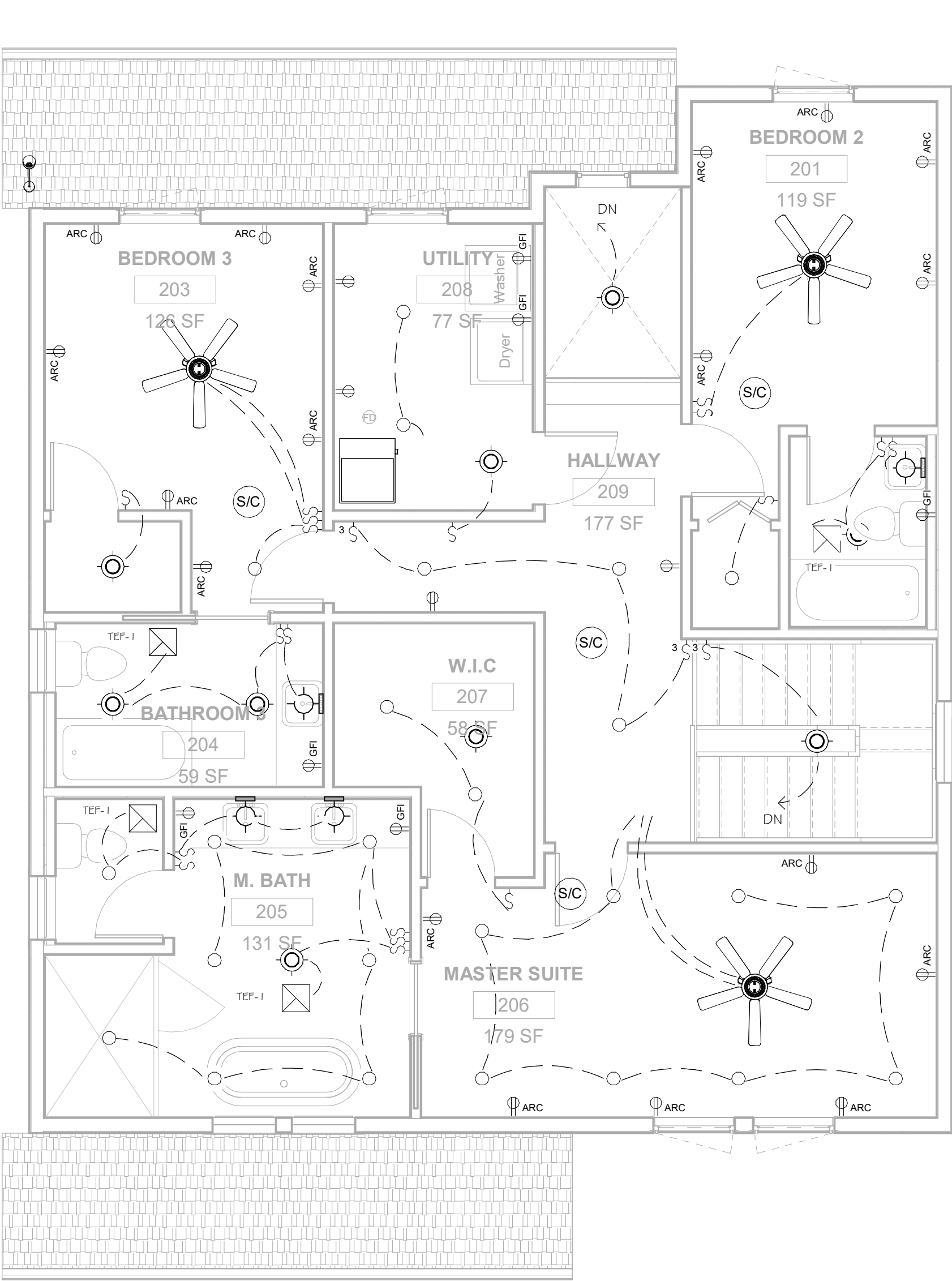
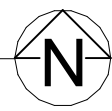
5. ALL MATERIALS USED IN THIS WORK SHALL BE NEW, OR/OF RECENT MANUFACTURE AND BEAR THE UNDERWRITERS LABORATORIES LABEL.
6. THE ELECTRICAL CONTRACTOR SHALL GUARANTEE ALL MATERIALS AND WORKMANSHIP FOR A PERIOD OF ONE YEAR FROM DATE OF FINAL ACCEPTANCE. HE SHALL REPAIR OR REPLACE AT HIS OWN EXPENSE ANY PART OF THE ELECTRICAL INSTALLATION WHICH MAY BECOME DEFECTIVE WITHIN THAT TIME, NOT DUE TO CARELESSNESS OF OPERATION.
7. THE ELECTRICAL CONTRACTOR SHALL VISIT THE SITE DURING THE BIDDING PERIOD AND FAMILIARIZE HIMSELF WITH ALL CONDITIONS EFFECTING HIS WORK. LATER CLAIMS FOR LABOR OR MATERIALS REQUIRED FOR DIFFICULTIES ENCOUNTERED WHICH COULD HAVE BEEN FORESEEN BY SUCH AN INSPECTION WILL NOT BE RECOGNIZED.
8. THE ELECTRICAL CONTRACTOR SHALL COORDINATE HIS WORK WITH ALL TRADES ENGAGED ON THIS PROJECT. HE SHALL COOPERATE WITH ALL TRADES IN ORDER TO COORDINATE ALL WORK AND ELIMINATE CONFLICTS BETWEEN HIS WORK AND THAT OF OTHER TRADES.
9. THE ELECTRICAL CONTRACTOR SHALL PROVIDE TEMPORARY POWER AND LIGHTING DURING THE ENTIRE CONSTRUCTION PERIOD, AS REQUIRED FOR ALL TRADES.
10. THE ELECTRICAL CONTRACTOR SHALL PREPARE A SET OF " AS - BUILT " REPRODUCIBLE DRAWINGS WHICH SHALL SHOW THE ENTIRE INSTALLATION, INCLUDING CONDUIT RUNS, EQUIPMENT LOCATIONS AND ALL CHANGES TO THE ORIGINAL CONTRACT DOCUMENTS.
11. ALL WIRE SHALL BE COOPER. WIRE INSULATION FOR DRY LOCATIONS SHALL BE TYPE THHN AND TYPE THWN FOR WET LOCATIONS. THE MINIMUM CONDUCTOR SIZE SHALL BE #12 EXCEPT FOR CONTROL AND SIGNAL CABLES.
12. ALL WIRE #8 AND SMALLER SHALL BE SOLID AND ALL WIRE #6 AND LARGER SHALL BE STRANDED.
13. EQUIPMENT INSTALLED IN CEILING AREAS USED AS AN AIR PLENUM IS TO BE APPROVED FOR USE IN SUCH AN AREA. VERIFY THE LIMITS OF ALL SUCH AREAS WITH THE MECHANICAL DRAWINGS.
14. ALL REQUIRED EXCAVATION AND BACKFILL FOR THE INSTALLATION OF ELECTRICAL WORK SHALL BE INCLUDED IN THE ELECTRICAL CONTRACT. ALL CUTTING, DRILLING, CHANNELING, PATCHING AND SLEEVES REQUIRED FOR THE INSTALLATION OF THE ELECTRICAL WORK SHALL BE INCLUDED IN THE ELECTRICAL CONTRACT. BEFORE ANY STRUCTURAL WORK IS CUT, DRILLED OR OTHERWISE MODIFIED APPROVAL MUST BE OBTAINED FROM THE ARCHITECT.
15. ALL FUSES SHALL BE FUSETRON OR APPROVED EQUAL.
16. ALL CONDUIT INSTALLED IN UNFINISHED AREAS SHALL BE EXPOSED AND ALL WORK IN FINISHED AREAS SHALL BE CONCEALED. CONTRACTOR SHALL NOT INSTALL CONDUITS IN SLABS ON GRADE. CONDUITS SHALL BE INSTALLED 4" BELOW THE BOTTOM OF THE CONCRETE SLAB SO THAT THE SLABS CAN BE REPLACED. GALVANIZED RIGID STEEL CONDUIT SHALL BE USED FOR ALL UNDERGROUND WORK AND FOR SERVICE ENTRANCE CONDUCTORS. ELECTRICAL METALLIC TUBING SHALL BE USED FOR ALL OTHER AREAS WHERE ALLOWED BY CODE. ALL EMT SHALL USE GLAND TYPE COMPRESSION FITTINGS.
17. RIGID NON-METALLIC PVC SCHEDULE 40 SHALL BE INSTALLED FOR ALL EXTERIOR UNDERGROUND WIRING WHERE ALLOWED BY CODE. IF NOT ALLOWED BY CODE GALVANIZED RIGID STEEL CONDUIT SHALL BE INSTALLED. FOR PVC RACEWAY SYSTEMS ALL ELBOWS AND RISERS EMERGING FROM GRADE, THE CABLE SHALL BE ALLOWED ONLY IN UNPAVED AREAS WHERE DIRECT BURIAL CABLE IS SPECIFICALLY INDICATED AND SHOWN ON THE PLANS.
18. LIGHT SWITCHES SHALL BE HUBBELL 1221-1, 1223-1 AND 1224-1. CONVENIENCE OUTLETS SHALL BE HUBBELL 5252-1. ALL SWITCH AND RECEPTACLE PLATES FOR CONCEALED WORK SHALL BE SMOOTH IVORY PLASTIC. PLATES FOR EXPOSED WORK SHALL BE GALVANIZED PRESSED STEEL. WHERE GROUPED SWITCHES ARE SHOWN, GONG PLATES SHALL BE USED. CONVENIENCE OUTLETS FOR FLOOR INSTALLATION SHALL BE HUBBELL LO-CON.
19. THE MOUNTING HEIGHTS OF EQUIPMENT SHALL BE AS FOLLOWS UNLESS OTHERWISE NOTED ON THE DRAWINGS: LIGHT SWITCHES.....4' 6" TO CENTERLINE RECEPTACLES.....4' 0" IN UNFINISHED AREAS 1' 6" IN FINISHED AREAS
20. DEFINITIONS: PROVIDE: TO FURNISH, INSTALL AND CONNECT COMPLETE AND READY FOR OPERATION OF THE PARTICULAR WORK REFERRED TO, UNLESS SPECIFICALLY NOTED OTHERWISE. INSTALL: TO ERECT, MOUNT AND CONNECT COMPLETE WITH RELATED ACCESSORIES. FURNISH: TO PURCHASE, PROCURE, ACQUIRE AND DELIVER COMPLETE WITH RELATED ACCESSORIES. WORK: LABOR, MATERIALS, EQUIPMENT, APPARATUS, CONTROLS, ACCESSORIES, AND OTHER ITEMS REQUIRED FOR A COMPLETE INSTALLATION.
21. LIGHTING PANELS ARE TO BE CIRCUIT BREAKER TYPE WITH BOLT-ON TYPE BREAKERS, DOORS, KEYED LOCKS, FLUSH HANDLES, FLUSH TRIM AND CIRCUIT DIRECTORY; SQUARE D TYPE NQOD OR APPROVED EQUAL. PANELS SHALL HAVE AN INTEGRATED SHSOKT CIRCUIT RATING SUITABLE FOR THE MAXIMUM AVAILABLE FAULT CURRENT AVAILABLE AT THE PANEL LOCATION AS NOTED ON THE DRAWING, WHICHEVER IS THE LARGEST. PROVIDE TYPEWRITTEN DIRECTORIES FOR ALL PANELBOARDS SHOWING FINAL CIRCUITING.



1 ELECTRICAL FIRST FLOOR
1/4" = 1'-0"



2 ELECRICAL SECOND FLOOR
1/4" = 1'-0"



LOAD CALCULATION 2018 ELECTRICAL CODE										
					DATE:		2/12/26			
PROJECT NAME:					NEW CONSTRUCTION OF A SFR					
ADDRESS:					3521 134th PI Robbins, IL 60472					
BLDG. EXTERIOR DIMENSIONS					LENGTH:		38.00		WIDTH:	
TOTAL No. OF FLOORS:					3		TOTAL No. OF UNITS/APTS.:		3	
TYPE OF SERVICE:					480. Y/277V. 3ø. 4W		208/120V. 3ø. 4W		120/208V. 1ø. 3W	
					480V. 3ø. 3W		208V. 3ø. 3W		X 120/240V. 1ø. 3W	
					* (REQUIRES GROUND DETECTION)					
SERVICE OCP:					200A					
SERVICE FEEDER:					3-3/ø					
GENERAL LIGHTING LOAD 220.16										
SQ. FT. PER FLOOR:					1334		X No. OF FLOORS		2	
APPLIANCE CKTS/APT*:					1		x 1500 WATTS/CKT. x No. OF APTS		x3 WATTS/SQ. FT.	
LAUNDRY CKTS/APT:					1		x 1500 WATTS/CKT. x No. OF APTS		1	
*(MINIMUM 2 CIRCUITS/APT)									=	
									8,004	
									1500	
									1500	
									=	
									11,004	
									WATT \$	
DEMAND 220.12										
FIRST 20,000 WATTS OF GEN. LTG. LOAD @ 100%									=	
NEXT 100,000 WATTS OF GEN. LTG. LOAD @ 35%									11,004.00	
ALL IN EXCESS OF 120,000 WATTS @ 25%									=	
									0	
									0	
									=	
									11,004.00	
									WATT \$	
APPLIANCE LOADS 220.17, 220.18, 220.19										
					0		PER TABLE 19-27-220 19 COLUMN A			
No. OF ELECTRIC RANGES/OVENS					0		8000		WATTS EA. @	
No. OF GARBAGE DISPOSALS					1		1200		100% DEMAND=	
No. OF DISHWASHERS					1		1500		75% DEMAND=	
No. OF ELEC. CLOTHES DRYERS					0		5,000		75% DEMAND=	
No. OF ELEC. HOT WATER HTR					0		4000		100% DEMAND=	
No. OF FAN COIL MOTORS (AIR HANDLER)					1		722		75% DEMAND=	
No. OF ELEC. HTG. COILS (ELECTRIC FURNACE)					1		722		100% DEMAND=	
No. OF ELEC. BASEBOARD HTRS					0		15000		100% DEMAND=	
No. OF AIR-CONDITIONERS					0		1500		100% DEMAND=	
No. OF LARGEST MOTOR					1		4000		100% DEMAND=	
No. OF REFRIGERATORS					0		4080		100% DEMAND=	
No. OF MICROWAVES					1		1000		25% DEMAND=	
No. OF MISC. APPLIANCES (EV READY CHARGER)					1		1000		75% DEMAND=	
					0		12000		75% DEMAND=	
									0	
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MECHANICAL GENERAL NOTES

1. MECHANICAL SYSTEM SHALL BE INSTALLED IN ACCORDANCE WITH CITY OF CHICAGO, ILLINOIS BUILDING CODES.
2. THE MECHANICAL CONTRACTOR SHALL COORDINATE DUCTWORK INSTALLATION WITH ARCHITECTS/OWNER'S REPRESENTATIVE AND ALL OTHER TRADES THROUGH A COORDINATED SHOP DRAWING REVIEW PROCESS. SEE ARCHITECTURAL DRAWINGS FOR EXACT LOCATION AND HEIGHTS.
3. CORE-DRILL OR SAW-CUT OR SLEEVE FLOOR WALL, ROOF, ETC. AS REQUIRED FOR PIPING OR DUCTWORK AND FIRE-STOP OPENING AROUND PIPE AND DUCTWORK. VERIFY LOCATION OF STRUCTURAL BEAMS, JOISTS, ETC. BEFORE DRILLING OR CUTTING. NOTIFY ARCHITECT OF ANY DISCREPANCIES.
4. THE CONTRACTOR SHALL WARRANTEE ALL MATERIAL AND GUARANTEE ALL WORKMANSHIP FOR ONE YEAR FROM SUBSTANTIAL COMPLETION.
5. ALL CONTRACTOR FABRICATED AND MANUFACTURER FABRICATED COMPONENTS OF THE OUTSIDE AIR, SUPPLY AIR, RETURNING AND EXHAUST SYSTEMS SHALL BE CONSTRUCTED AND INSTALLED AIR-TIGHT. PIPE OPENINGS IN SYSTEM COMPONENT SHALL HAVE SHEET METAL BAFFLES, SET IN SEALANT, TO PREVENT LEAKAGE.
6. THE MECHANICAL CONTRACTOR SHALL INSTALL MECHANICAL SYSTEMS AS SHOWN, NOTED AND SPECIFIED. EQUIPMENT MAY NOT BE SUBSTITUTED UNLESS WRITTEN APPROVAL BY THE ARCHITECT, ENGINEER, OR OWNER'S REPRESENTATIVE IS OBTAINED. ANY UNAUTHORIZED CHANGES WILL BE REMOVED AT CONTRACTOR'S EXPENSE, IF DEEMED NECESSARY BY ARCHITECT, ENGINEER, OR OWNER'S REPRESENTATIVE.
7. DRAWINGS FOR MECHANICAL WORK ARE DIAGRAMATIC, SHOWING THE GENERAL LOCATION, TYPE, LAYOUT AND EQUIPMENT REQUIRED. THE DRAWINGS SHALL NOT BE SCALED FOR EXACT MEASUREMENTS. REFER TO MANUFACTURER'S STANDARD INSTALLATION DRAWINGS FOR EQUIPMENT CONNECTIONS AND INSTALLATION REQUIREMENT AS REQUIRED. FURNISH AND INSTALL DUCTWORK, CONNECTIONS, ACCESSORIES, OFFSETS AND MATERIALS NECESSARY TO FACILITATE THE SYSTEM'S FUNCTIONING AS INDICATED BY THE DESIGN AND THE EQUIPMENT INDICATED. THE WORK SHALL BE IN ACCORDANCE WITH LOCAL CODES AND ORDINANCES AND SUBJECT TO INSPECTION.
8. ALL DUCTWORK AND PIPING SHALL BE ROUTED ABOVE THE SUSPENDED CEILING SPACE, UNLESS OTHERWISE NOTED.
9. ALL DUCTWORK IN LIVING UNITS SHALL BE ROUTED IN CONCEALED SPACES OR SOFFITS WITHOUT INSULATION.
10. ALL DUCTWORK AND PIPING THAT IS EXPOSED TO VIEW SHALL BE ROUTED AS HIGH AS POSSIBLE AND TIGHT TO THE UNDERSIDE OF THE STRUCTURE ABOVE, UNLESS OTHERWISE NOTED.
11. ALL DUCTWORK SHALL BE CONSTRUCTED OF GALVANIED IRON SHEET METAL AND BE FABRICATED ACCORDING TO THE S.M.A.C.N.A. LOW VELOCITY DUCT MANUAL, ASHRAE 1996 HANDBOOK VOLUME "HVAC SYSTEMS AND EQUIPMENT". ALL ELBOWS SHALL HAVE PROPER RADIUS, OR MECHANICAL CONTRACTOR SHALL PROVIDE DOUBLE THICKNESS, AIRFOIL TURNING VANES REQUIRED BY S.M.A.C.N.A. NO SQUARE THROAT ELBOWS SHALL BE INSTALLED WITHOUT DOUBLE THICKNESS TURNING VANES.
12. ALL BRANCH SUPPLY DUCTS SHALL HAVE (VD) MANUAL VOLUME DAMPERS INSTALLED FOR BALANCING.
13. PROVIDE A VOLUME DAMPER FOR EVERY INLET AND OUTLET (DIFFUSERS, REGISTERS, GRILLES, ETC.) OF THE DUCTWORK DISTRIBUTION SYSTEMS WHETHER SHOWN OR NOT ON THE PLANS, PROVIDE ADDITIONAL VOLUME DAMPERS OR EXTRACTORS AT BRANCH TAKE-OFFS FROM DUCTWORK MAINS AS REQUIRED TO ACHIEVE AIR VOLUME DISTRIBUTION AND BALANCING. RESIDENTIAL FAN COIL UNIT AIR BALANCING DAMPERS ARE NOT REQUIRED.
14. ALL RECTANGULAR/SQUARE DUCTWORK TRANSITIONS SHALL BE (FOT) "FLAT ON TOP", UNLESS OTHERWISE SPECIFIED ON PLAN.
15. ALL DUCT CONNECTIONS TO EQUIPMENT SHALL BE MADE WITH NEOPRENE, DOUBLE COATED, HEAVY GLASS FABRIC, VIBRATION ELIMINATION CONNECTIONS, (F.C.) FLEXIBLE CONNECTIONS, EQUAL TO VENTFABRICS, INC. TYPE VENTGLAS, UNLESS NOTED OTHERWISE.
16. OUTSIDE AIR INTAKES SHALL BE A MINIMUM OF 1'-0" AWAY FROM EXHAUST DISCHARGE OPENINGS AND PLUMBING VENT STACKS.
17. SUPPLY, RETURN AND OUTSIDE AIR RECTANGULAR/SQUARE DUCTWORK SHALL BE INSULATED WITH 1-1/2" THICK, 3 PCF DUCT LINER. OUTDOORS DUCTWORK SHALL BE INSULATED WITH 2" THICK 6 PCF DUCT LINER. DUCT LINER SHALL BE FASTENED TO INSIDE OF DUCTWORK AS PER MANUFACTURER'S DIRECTIONS AND S.M.A.C.N.A. "DUCT LINER APPLICATION STANDARD". SIZE OF DUCTS SHALL BE INCREASED FOR DUCT LINER. INSULATION. SIZES SHOWN ON PLAN ARE INSIDE FREE AREA. ROUND SUPPLY AND RETURN DUCTWORK SHALL BE INSULATED WITH 2" THICK, 1-PCF FIBERGLASS DUCTWRAP WITH FSK FOIL VAPOR BARRIER. ROUND SUPPLY AND RETURN DUCTWORK EXPOSED TO VIEW AND LOCATED IN THE CONDITIONED SPACE SHALL NOT BE INSULATED.
18. THE MECHANICAL SYSTEM SHALL BE COMPLETE WITH ALL NECESSARY APPURTENANCES FOR A COMPLETE OPERATING SYSTEM.
19. MECHANICAL CONTRACTOR SHALL FURNISH AND INSTALL ALL CONTROL WIRING AS REQUIRED. THERMOSTATS AND HUMIDISTATS AND SENSORS SHALL BE AS SPECIFIED, OR AS FURNISHED WITH THE EQUIPMENT. PROVIDE TRANSFORMERS AS REQUIRED.
20. PROVIDE UL APPROVED FIRE DAMPERS FOR ALL PENETRATIONS THROUGH FIRE RATED WALLS, PARTITIONS, CEILINGS, AND FLOORS. INSTALL FIRE DAMPERS AS PER MANUFACTURER'S DIRECTIONS AND AS PER UL GUIDE LINES. PROVIDE ACCESS PANELS IN BUILDING CONSTRUCTION AS REQUIRED FOR SERVICING OF FIRE DAMPERS.
21. BALANCING CONTRACTOR SHALL BALANCE SYSTEM TO AIR QUANTITIES SHOWN ON PLAN. BALANCING CONTRACTOR SHALL USE DUCT MOUNTED MANUAL DAMPERS FOR AIR SYSTEM BALANCING. USE OF TERMINAL DAMPERS IS NOT ACCEPTABLE. RESIDENTIAL FAN COIL UNIT AIR BALANCE IS NOT REQUIRED.
22. PROVIDE A COMPLETE NATURAL GAS PIPING SYSTEM AS NOTED ON THE DRAWINGS. PROVIDE EITHER TREADED STEEL OR MALLEABLE IRON PIPE WITH MALLEABLE FITTINGS OR WELDED STEEL. PROVIDE ALL UNIONS, SHUT-OFF VALVES AND DIRT LEGS REQUIRED BY NFPA-54 AND GOVERNING LOCAL CODES. PROVIDE ALL TESTS, METERS, INSPECTIONS, HANGERS AND EQUIPMENT CONNECTIONS REQUIRED FOR A COMPLETE AND OPERATING SYSTEM. PAINT PIPING ON ROOF WITH TWO COATS OF RUST RESISTANT OUTDOOR PAINT.
23. VENTING OF ALL GAS FIRED APPLIANCES SHALL CONFORM TO INTERNATIONAL FUEL GAS CODE.
24. FUEL GAS PIPING AND CONTROLS SHALL CONFORM TO INTERNATIONAL FUEL GAS CODE.
25. GAS PIPING MATERIALS SHALL CONFORM TO INTERNATIONAL FUEL GAS CODE.
26. PORTIONS OF GAS PIPING SYSTEM INSTALLED IN CONCEALED LOCATIONS SHALL NOT HAVE UNIONS, TUBING FITTINGS, RIGHT AND LEFT COUPLINGS, BUSHINGS, COMPRESSION COUPLINGS AND SWING JOINTS MADE BY COMBINATIONS OF FITTINGS, EXCEPT TUBING JOINED BY BRAZING OR FITTINGS LISTED FOR USE IN CONCEALED LOCATIONS.
27. GAS PIPING SHALL BE SLOPED NOT LESS THAN 1/4 INCH IN 15 FEET TO PREVENT TRAPS.
28. DUCT SYSTEMS DESIGNED AND SIZED PER ASHRAE LOAD CALCULATION STANDARDS.
29. PERFORM DUCT LEAKAGE TEST TO CONFIRM THAT DUCT LEAKAGE IS LESS THAN 10% OF THE MAXIMUM TOTAL SYSTEM AIR FLOW.

GAS PIPING NOTES

1. GAS PIPING FROM INCOMING SERVICE PRESSURE REGULATOR TO THE GAS APPLIANCES AND EQUIPMENT, INCLUDING VALVES, UNIONS, METERS SPREADS, SUPPORTS, HANGERS, SLEEVES, ETC. SHALL BE PROVIDED BY THE CONTRACTOR. INCOMING SERVICE WITH SHUT-OFF VALVE, PRESSURE REGULATOR AND METERS, BY THE GAS COMPANY.
2. DRAWING FOR GAS PIPING WORK IS DIAGRAMMATIC, SHOWING THE GENERAL LOCATION, TYPE, LAYOUT AND EQUIPMENT REQUIRED. THE DRAWINGS SHALL NOT BE SCALED FOR EXACT MEASUREMENTS. REFER TO MANUFACTURER'S STANDARD INSTALLATION DRAWINGS FOR EQUIPMENT CONNECTIONS AND INSTALLATION REQUIREMENTS AS REQUIRED. FURNISH AND INSTALL PIPING, CONNECTIONS, ACCESSORIES, OFFSETS AND MATERIALS NECESSARY TO FACILITATE THE SYSTEM FUNCTIONING AS INDICATED BY THE DESIGN AND THE EQUIPMENT INDICATED. THE WORK SHALL BE IN ACCORDANCE WITH LOCAL CODES AND ORDINANCES, GAS COMPANY REQUIREMENTS AND SUBJECT TO INSPECTION.
3. PORTIONS OF A PIPING SYSTEM INSTALLED IN CONCEALED LOCATIONS SHALL NOT HAVE UNIONS, TUBING FITTINGS, RIGHT AND LEFT COUPLINGS, BUSHINGS, COMPRESSION COUPLINGS AND SWING JOINTS MADE BY COMBINATIONS OF FITTINGS EXCEPT TUBING THAT IS JOINED BY BRAZING OR FITTINGS LISTED FOR USE IN CONCEALED LOCATIONS.
4. GAS PIPES MUST BE SLOPED AT 1/4" PER 15 FT.

5. EACH APPLIANCE SHALL BE PROVIDED WITH SHUT-OFF VALVE SEPARATE FROM THE APPLIANCE. THE SHUT-OFF VALVE SHALL BE LOCATED IN THE SAME ROOM AS THE APPLIANCE, NOT FURTHER THAN 6 FT FROM THE APPLIANCE, AND SHALL BE INSTALLED UPSTREAM FROM THE UNION, CONNECTOR OR QUICK DISCONNECT DEVICE IT SERVES, SUCH SHUT-OFF VALVES SHALL BE PROVIDED WITH READY ACCESS. EQUIPMENT SHUT-OFF VALVES LOCATED IN THE FIREBOX OF A FIREPLACE SHALL BE INSTALLED IN ACCORDANCE WITH THE APPLIANCE MANUFACTURER'S INSTRUCTIONS.

NATURAL GAS DISTRIBUTION PIPING SHALL BE SCHEDULE 40 BLACK PIPE WITH STANDARD MALLEABLE IRON FITTINGS. GAS PIPING GREATER THAN 2" INSIDE DIAMETER OR CARRYING MORE THAN 5 POUNDS(PSIG) SHALL BE SCHEDULE 40 STANDARD WELD FITTINGS.

HVAC SPECIFICATIONS

DRAWINGS AND GENERAL PROVISIONS THE CONTRACT, INCLUDING GENERAL AND SUPPLEMENTARY CONDITIONS AND DIVISION 1 SPECIFICATION SECTIONS, APPLY TO THIS SECTION.

ALL MECHANICAL WORK SHALL BE INSTALLED IN STRICT ACCORDANCE WITH THE CHICAGO CONSTRUCTION CODE 2019.

HVAC WORK SHALL BE AS INDICATED ON THE PLANS AND AS HEREIN SPECIFIED. THE WORK SHALL CONSIST OF PROVIDING COMPLETE OPERATIONAL HVAC SYSTEMS AS INDICATED ON THE DRAWINGS. THE SYSTEM SHALL BE TESTED AND BALANCED. CAPACITIES SHALL BE AS SHOWN ON THE WITHIN THE INTENT OF THE DRAWINGS AND SPECIFICATIONS AND TO PROVIDE AN ACCEPTABLE SYSTEM IN ACCORDANCE WITH THE STATE CODES, LOCAL CODES, ASHRAE AND SMACNA RECOMMENDATIONS.

WHENEVER A MATERIAL, ARTICLE, OR PIECE OF EQUIPMENT IS IDENTIFIED ON THE DRAWINGS BY REFERENCE TO MANUFACTURERS OR VENDORS NAMES, TRADE NAMES, CATALOG NUMBERS, OR THE MATERIAL, ARTICLE, OR PIECE OF EQUIPMENT OF OTHER MANUFACTURERS OR VENDORS WHICH WILL PERFORM ADEQUATELY THE DUTIES IMPOSED BY THE GENERAL DESIGN WILL BE CONSIDERED EQUALLY ACCEPTABLE PROVIDED THE MATERIAL, ARTICLE, OR PIECE OF EQUIPMENT SO PROPOSED IS OF EQUAL SUBSTANCE, APPEARANCE, AND FUNCTION.

DUCTWORK:

DUCTWORK SHALL BE OF THE SIZE AS INDICATED ON THE PLANS WITH INTERNAL INSULATION PROVIDED ON ALL RECTANGULAR SUPPLY AND RETURN DUCTWORK. INSULATION SHALL BE 1" THICK 1 - 1/2 PCF FIBERGLASS, NEOPRENE COATED, AND ADHERED WITH AN APPROVED ADHESIVE WITH 100% COVERAGE AND STICK CLIPS ON 12" CENTERS. ROUND DUCTWORK, BOTH METAL AND FLEXIBLE, SHALL BE INSULATED WITH 1" THICK, 1 - 1/2 PCF FIBERGLASS INSULATION. ALL DUCTWORK SHALL BE FABRICATED AND INSTALLED IN ACCORDANCE WITH ASHRAE AND SMACNA RECOMMENDATIONS. ALL DUCTWORK SIZES INDICATED ARE NET SIZES. DUCTWORK SHALL BE INCREASED TO ALLOW FOR INTERNAL INSULATION. ALL SQUARE ELBOWS SHALL INCLUDE SINGLE THICKNESS TURNING VANES.

EQUIPMENT:

ALL AIR HANDLING EQUIPMENT, DIFFUSERS, GRILLES, REGISTERS, FANS, REFRIGERATION UNITS, ETC., SHALL BE PROVIDED AS SCHEDULED ON THE DRAWINGS. CONTROLS SHALL BE PROVIDED AS REQUIRED AND DESCRIBED ON THE DRAWINGS AND IN ACCORDANCE WITH NEC REQUIREMENTS. ALL RELAYS, INTERLOCKS, ETC., SHALL BE PROVIDED AS REQUIRED. POWER WIRING SHALL BE PROVIDED BY THE ELECTRICAL CONTRACTOR UP TO AND INCLUDING FINAL CONNECTION. THE MECHANICAL CODE UNLESS INDICATED TO BE PROVIDED BY THE ELECTRICAL CONTRACTOR. THE MECHANICAL CONTRACTOR SHALL MOUNT ALL DEVICES, ETC., AND THE ELECTRICAL CONTRACTOR SHALL WIRE THROUGH THEM AS REQUIRED. ALL ELECTRICAL WORK SHALL BE IN ACCORDANCE WITH THE NEC. REFRIGERANT PIPING WHERE REQUIRED SHALL BE OF THE PRECHARGED TYPE AS SUPPLIED BY THE FABRICATED AT THE CONTRACTOR'S OPTION. FIELD FABRICATED REFRIGERANT PIPING SHALL BE TYPE "L" A.R.I. COPPER, SOLDERED WITH 95/5 SOLDER. INSULATED WITH 1" ARMAFLEX AND WEATHERPROOFED. PIPING SHALL BE ARRANGED AND PIPED IN ACCORDANCE WITH THE EQUIPMENT MANUFACTURER'S RECOMMENDATIONS AND SHALL INCLUDE SIGHT GLASSES, SERVICE VALVES, AND FILTER DRYERS WITH BY-PASSES WHEN INDICATED OR AS RECOMMEND BY THE EQUIPMENT MANUFACTURER FOR PROPER EQUIPMENT OPERATION. ALL PIPING ARRANGEMENTS SHALL BE APPROVED BY BOTH THE EQUIPMENT MANUFACTURER AND THE ARCHITECT PRIOR TO INSTALLATION.THE CONTRACTOR SHALL TEST ALL DUCT AND PIPING SYSTEMS FOR LEAKAGE, BALANCE AIR AS INDICATED, CHARGE REFRIGERATION EQUIPMENT, PROVIDE CLEAN FILTERS, START SYSTEMS, AND INSTRUCT THE OWNER IN OPERATION OF THE SYSTEM AFTER ALL TEST AND BALANCING HAS BEEN ACCOMPLISHED. ALL LEAKS OBSERVED IN THE SYSTEM SHALL BE REPAIRED BY EITHER REPLACING OR REPAIRING THE DEFECTIVE MATERIAL OR EQUIPMENT IN A MANNER APPROVED BY THE ARCHITECT. THE ENTIRE HVAC SYSTEM SHALL BE UNCONDITIONALLY GUARANTEED FOR ONE YEAR FROM THE DATE OF ACCEPTANCE BY THE OWNER. ALL DEFECTIVE EQUIPMENT AND/OR MATERIAL FLEXIBLE CONNECTORS AT EQUIPMENT CONNECTIONS AND INSULATED DRAIN LINES PIPED TO FLOOR DRAINS, HUB DRAINS, OR OTHER APPROVED DRAIN POINTS.

GAS PIPING:

PIPE AND FITTING SHALL BE BLACK STEEL WITH MALLEABLE IRON FITTINGS UP THROUGH 2" I.P.S. AND STEEL WELDING FITTINGS 2 - 1/2" I.P.S. AND ABOVE. DIELECTRIC COUPLINGS OR UNIONS SHALL BE UTILIZED WHEN PIPING OF DISSIMILAR METAL IS CONNECTED.

FOR THE INSTALLATION OF THE GAS METER, GAS SERVICE, AND ITS ACCESSORIES. GAS PIPING SYSTEM SHALL BE INSTALLED TO THE REQUIREMENTS OF THE AGA PAMPHLET "INSTALLATION OF GAS APPLIANCES AND GAS PIPING" AND NFPA STANDARD NO. 54. THE GAS PIPING SHALL BE TESTED IN ACCORDANCE WITH THE PROCEDURES DESCRIBED IN NFPA NO. 54. ANY OTHER TESTS AS REQUIRED BY THE LOCAL BUILDING INSUAL BUILDING INSPECTION DEPARTMENT OR GAS COMPANY SHALL ALSO BE PERFORMED. PAINT GAS PIPING WITH RUST RESISTANT PAINT.

MECHANICAL ABBREVIATIONS

ACCU	AIR COOLED CONDENSING UNIT
DEF	DRYER EXHAUST FAN
ECUH	ELECTRIC CABINET UNIT HEATER
EDH	ELECTRIC DUCT HEATER
EHUH	ELECTRIC HORIZONTAL UNIT HEATER
EF	EXHAUST FAN
FCU	FAN COIL UNIT
FD	FIRE DAMPER
L	LOUVER
MAU	MAKE-UP AIR UNIT
MOD	MOTOR OPERATED DAMPER
P	PUMP
RTU	ROOFTOP UNIT
SF	SUPPLY FAN
TEF	TOILET EXHAUST FAN
VRFFCU	VARIABLE REFRIGERANT FAN COIL UNIT
VRPHP	VARIABLE REFRIGERANT HEAT PUMP
AD	ACCESS DOOR
AFB	ABOVE FINISHED FLOOR
ARCHI	ARCHITECT OR ARCHITECTURAL
BLDG	BUILDING
BOT	BOTTOM
CLG	CEILING
COL	COLUMN
CON	CONCRETE
CONST'N	CONSTRUCTION
CONT'R	CONTRACTOR
DET	DETAIL
DN	DOWN
DWG	DRAWING
DG	DOOR GRILLE, SEE ARCH. DWGS.
DWFF	DWFF
(E)	EXISTING
EAT	ENTERING AIR TEMPERATURE
EWT	ENTERING WATER TEMPERATURE
FLR	FLOOR
FC	FLEXIBLE CONNECTION
FOB	FLAT ON BOTTOM
FOT	FLAT ON TOP
GA	GAUGE
KEF	KITCHEN HOOD EXHAUST FAN
KMAU	KITCHEN MAKE-UP AIR UNIT
LAT	LEAVING AIR TEMPERATURE
LWT	LEAVING WATER TEMPERATURE
MT'D	MOUNTED
NTS	NOT TO SCALE
OC	ON CENTER
(R)	RELOCATED
SS	STAINLESS STEEL
T	THERMOSTAT
TSP	TOTAL STATIC PRESSURE
TV	AIRFOIL TURNING VANES
UCD	UNDERCUT DOOR, SEE ARCH DWGS.
UNO	UNLESS NOTED OTHERWISE
W/	WITH
DIAM	DIAMETER/ROUND

LIGHT, VENTILATION AND HEAT LOSS SCHEDULE												
Level	Number	Name	Area	Natural Ventilation Per Code		Mechanical Ventilation Per Code		Actual Mechanical Ventilation		Actual BTU/HR Heat Loss	CFM to Heat - 1 OF to 70F	Actual Supply CFM
				Glass Req.	Vent Req.	Supply	Exhaust	Act Supply	Act Exhaust			
1st Floor	001	LIVING ROOM	247 SF	19.74	9.67	NV	NR	250	0	11106	199	250
1st Floor	002	FAMILY ROOM	200 SF	16.01	8.00	NV	NR	200	0	9003	162	200
1st Floor	003	DINING ROOM	160 SF	12.81	6.41	NV	NR	200	0	7206	129	200
1st Floor	004	KITCHEN	178 SF	14.21	7.10	NV	NR	200	300	7992	144	200
1st Floor	005	POWDER ROOM	30 SF	2.38	1.19	NV	NR	50	100	1340	24	50
1st Floor: 5										36646		900
2nd Floor	201	BEDROOM 2	119 SF	9.49	4.74	NV	NR	150	0	5337	96	150
2nd Floor	202	BATHROOM 2	35 SF	2.82	1.41	NV	NR	100	150	1565	28	100
2nd Floor	203	BEDROOM 3	126 SF	10.11	5.05	NV	NR	150	0	5686	102	150
2nd Floor	204	BATHROOM 3	59 SF	4.74	2.37	NV	NR	100	150	2666	48	100
2nd Floor	205	M. BATH	131 SF	10.44	5.22	NV	NR	150	200	5673	105	150
2nd Floor	206	MASTER SUITE	179 SF	14.29	7.15	NV	NR	200	0	8040	144	200
2nd Floor	207	W.I.C	58 SF	4.65	2.33	NV	NR	0	0	2618	47	0
2nd Floor	208	UTILITY	77 SF	6.18	3.09	NV	NR	50	100	3477	62	50
2nd Floor	209	HALLWAY	177 SF	14.18	7.09	NV	NR	200	0	7976	143	200
2nd Floor: 9										43259		1100

MECHANICAL EQUIPMENT SCHEDULE

NAME	DESCRIPTION	COUNT	LOCATION
GF-1	NATURAL GAS FIRED DOWNFLOW FURNACE. 120,000 BTU INPUT, 114,00 BTU OUTPUT. 95% EFFICIENCY "RHEEM" OR EQUAL	1	MECH CL.
TEF-1	TOILET EXHAUST FAN. 100 CFM. W/4" DUCT TO OUTSIDE, "BROAN" OR EQUAL	4	IN EACH W.C.

NOTE: CLEARANCES FOR FORCED AIR FURNACES MUST CONFORM TO MANUFACTURER'S REQUIREMENTS

REFRIGERATION SCHEDULE														
TAG	QUANTITY	SERVING	MANUF.	MODEL.	REMOTE	TONNAGE	REFR. TYPE	LOCATION	SELF-CONTAINED	REFER. WEIGHT	AIR COOLED	WATER COOLED	HP	REMAKRS
AC	1	ENTIRE BUILDING	RHEEM	N/A	YES	3 TON	410A	REAR YARD	NO	3 LB	YES	NO	1.5	



ARCHITECTONIC SOLUTIONS, INC.

4348 W LAKE ST STE 306
CHICAGO, IL 60624
312-373-0098

DESIGN FIRM NO. 184.006860-0001

I HEREBY CERTIFY THAT THESE PLANS WERE PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND TO THE BEST OF MY KNOWLEDGE AND BELIEF, THEY COMPLY WITH ALL OF ROBBINS, ILLINOIS BUILDING CODES AND ORDINANCES



SIGNATURE: *Oscar Alva*

EXPIRES: 11/30/26

	DATE	ISSUE
	11/21/25	FOR REVIEW
1	11/2/26	FIELD REVISION
2		
3		
4		
5		

JOB TITLE

NEW
2 STORY FRAME
RESIDENCE

PROJECT ADDRESS

3521 134th Pl.
Robbins. IL 60472

DRAWING TITLE

MECHANICAL
NOTES & DIAGRAMS

PROJECT # 3521

ISSUE DATE: 21/11/25

DRAWN BY: P.M.

CHECKED BY: O. ALBA

SHEET

M100

MECHANICAL PLAN LEGEND

SUPPLY
DIFFUSER

RETURN GRILL

TRANSFER GRILL

DAMPER

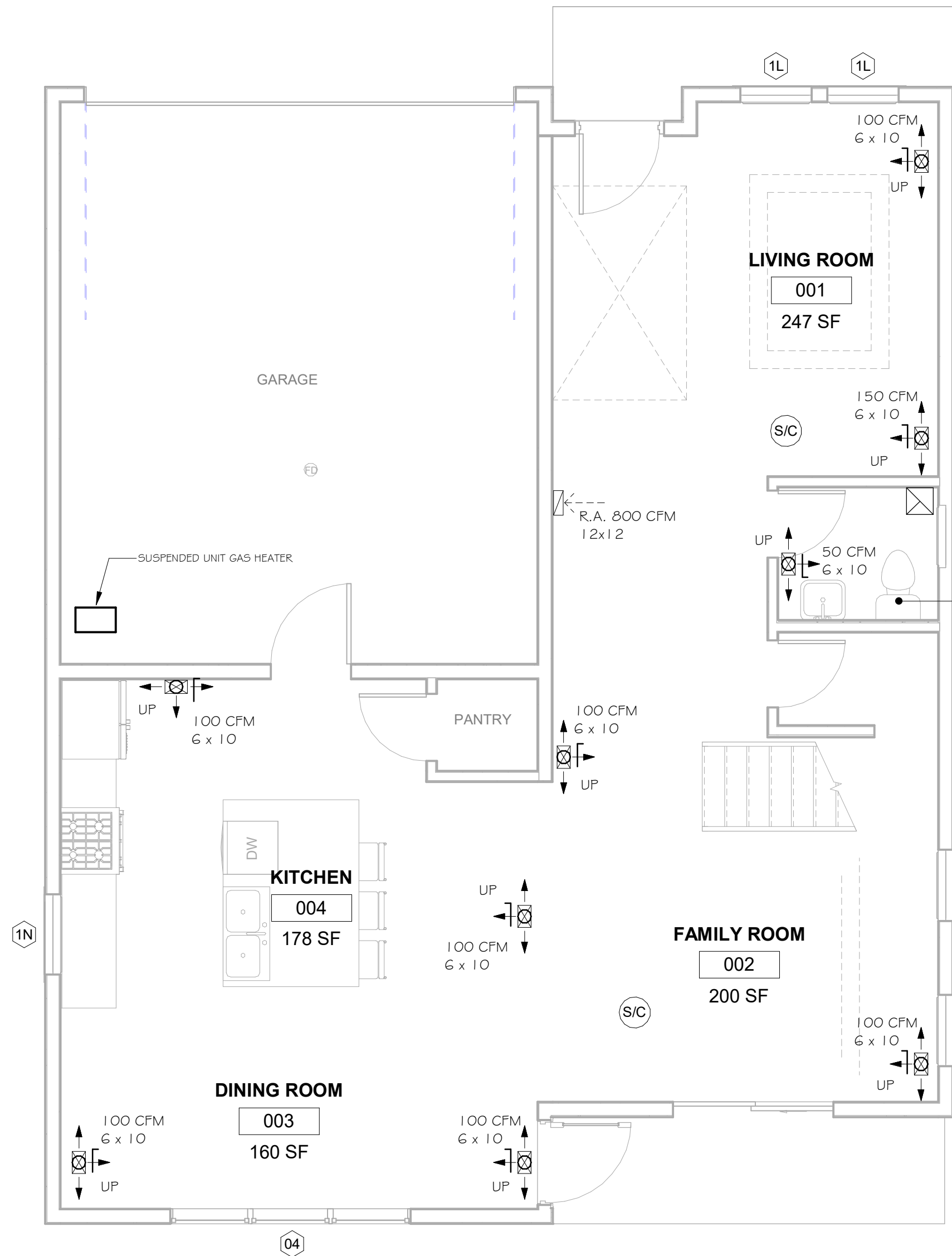
SUPPLY DUCT

RETURN DUCT

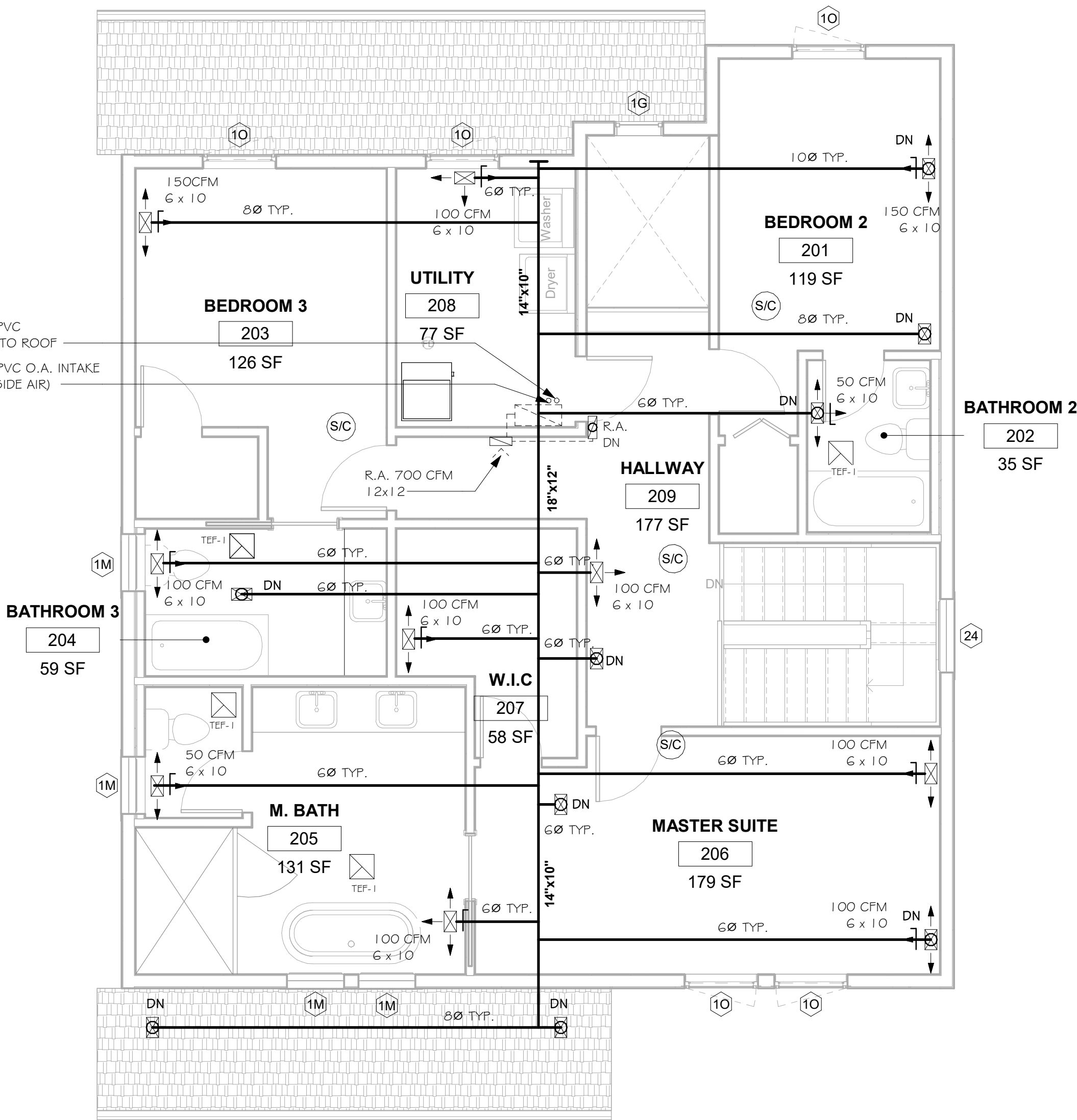
SMOKE DETECTOR &
CARBON MONOXIDE
HARD WIRED
INTERCONNECTED
W/BATTERY BACK UP

BATHROOM
EXHAUST FAN

WINDOW SCHEDULE							
Level	Mark	Count	Operation	Width	Height	Natural Light Area	Natural Vent Area
1st Floor	1L	4	Double Hung	2' - 8"	5' - 0"	11 SF	5.5 SF
1st Floor	1M	1	Double Hung	2' - 4"	3' - 6"	6 SF	3.5 SF
1st Floor	1N	1	Double Hung	3' - 0"	4' - 0"	9 SF	4.5 SF
1st Floor	04	1	Fixed	3' - 0"	5' - 6"	13 SF	0.0 SF
2nd Floor	1G	1	FIXED TRANSOM	2' - 0"	3' - 6"	5 SF	0.0 SF
2nd Floor	1M	4	Double Hung	2' - 4"	3' - 6"	6 SF	4.0 SF
2nd Floor	1O	5	CASEMENT	3' - 0"	4' - 5"	10 SF	5.0 SF
2nd Floor	24	1	Double Hung	3' - 0"	4' - 0"	9 SF	5.0 SF



1 MECHANICAL FIRST FLOOR
1/4" = 1'-0"



2 MECHANICAL SECOND FLOOR
1/4" = 1'-0"



ARCHITECTONIC
SOLUTIONS, INC.

4348 W LAKE ST STE 306
CHICAGO, IL 60624
312-373-0098

DESIGN FIRM NO. 184.006860-0001

I HEREBY CERTIFY THAT THESE
PLANS WERE PREPARED BY ME OR
UNDER MY DIRECT SUPERVISION AND
TO THE BEST OF MY KNOWLEDGE
AND BELIEF, THEY COMPLY WITH ALL
OF ROBBINS, ILLINOIS BUILDING
CODES AND ORDINANCES



SIGNATURE: *Oscar Alba*

EXPIRES: 11/30/26

DATE	ISSUE
11/21/25	FOR REVIEW
1/12/26	FIELD REVISION

JOB TITLE

NEW
2 STORY FRAME
RESIDENCE

PROJECT ADDRESS

352 I 134th Pl.
Robbins. IL 60472

DRAWING TITLE

MECHANICAL PLANS

PROJECT # 352 I

ISSUE DATE: 21/11/25

DRAWN BY: P.M.

CHECKED BY: O. ALBA

SHEET

M I I O

PLUMBING NOTES

1. THE PLUMBING CONTRACTOR SHALL INCLUDE ALL NECESSARY LABOR, MATERIALS AND EQUIPMENT REQUIRED FOR A COMPLETE AND OPERABLE INSTALLATION OF ALL WATER AND SEWAGE SYSTEM.

2. ALL SYSTEMS AND EQUIPMENT SHALL CONFORM TO APPLICABLE LOCAL, STATE AND NATIONAL CODES.

3. ALL PLUMBING PERMITS AND BONDS THAT ARE NOT TAKEN OUT WITH THE GENERAL BUILDING PERMIT SHALL BE TAKEN OUT AND PAID FOR BY THE PLUMBING CONTRACTOR.

4. THE PLUMBING CONTRACTOR SHALL PROVIDE A TEMPORARY WATER SUPPLY FOR USE DURING BUILDING CONSTRUCTION, PROVIDE PROTECTION FROM FREEZING.

5. SUSPENDED WATER PIPING SHALL BE TYPE "M" COPPER WITH NO LEAD OR 95-5 RATIO. INSULATE WHERE PIPING IS SUBJET TO FREEZING.

6. PLUMBING CONTRACTOR SHALL GUARANTEE ALL WORK AND MATERIAL FOR ONE YEAR AFTER COMPLETION AGAINST ALL DEFECTS OF MATERIALS, EQUIPMENT AND WORKMANSHIP.

7. IF LAUNDRY ON PLANS PROVIDE GREYBOX WITH SILL FAUCETS AND ONE STANDPIPE FOR WASHING MACHINE AND 2" DRAIN.

8. PROVIDE 12" AIR CHAMBER AT EVERY FIXTURE AND A 24" AIR CHAMBER AT TOP OF ALL MAIN WATER RISERS.

9. PROVIDE CHROME PLATED FIXTURE SUPPLIES AND STOPS FOR EVERY FIXTURE CONNECTION.

10. PROVIDE CLEAN-OUTS IN WASTE AND SOIL LINE FOR EACH CHANGE OF DIRECTION GREATER THAN 45 DEGREES AND AT THE BOTTOM OF EACH STACK.

11. ALL WATER PIPES SHALL BE VALVED.

12. WATER CLOSETS: 3-1/2 GALLON PER FLUSH MAXIMUM; SHOWERS, LAVS AND SINKS; 3 GALLON PER MINUTE FLOW CONTROL.

13. ALL FIXTURES TO HAVE VACUUM BREAKERS.

14. ALL VENT/WASTE PIPE TO BE PVC SCHEDULE 40

15. ALL COPPER ABOVE GROUND TO BE "L" TYPE

16. ALL COPPER BELOW GROUND TO BE "K" TYPE

17. ALL COPPER FITTINGS TO BE FORDGED WITH NONE LEAD SOLDER

18. NO LEAD SOLDER/FLUX

19. ALL PVC TO BE SCH. 40 TO MEET CODE

20. ALL PVC FITTINGS TO BE DWV WITH PURPLE

21. PRIMER AND CLEAR PVC CEMENT

22. SHUT OFF STOPS ON ALL FIXTURES

23. PLUMBER TO VERIFY THAT A 120 DEGREE WATER IS BEING SUPPLIED BY THE WATER HEATER TO THE KITCHEN SINK

24. PLUMBING CONTRACTOR TO TEST SYSTEM PRIOR TO OPERATION.

GENERAL NOTE 1 :

THIS INFORMATION SHOWN IN THESE DRAWINGS ARE IN SCHEMATIC FORM FOR PERMIT ISSUANCE ONLY. THE GENERAL CONTRACTOR AND SUBCONTRACTORS SHALL PROVIDE SHOP DRAWINGS COORDINATED WITH ARCHITECTURAL, MECHANICAL, PLUMBING, ELECTRICAL, AND STRUCTURAL DESIGN FOR REVIEW AND APPROVAL BY THE ARCH./ENG. OF RECORDS.

PIPING MATERIALS

WATER SUPPLY PIPING

1. WITH APPLICABLE PORTION OF CHICAGO CONSTRUCTION CODE, PLUMBING SECTION, PERTAINING TO PLUMBING MATERIALS, CONSTRUCTION, AND INSTALLATION OF PRODUCTS.

2. COMPLY WITH APPLICABLE AMERICAN NATIONAL STANDARDS PERTAINING TO PRODUCTS AND INSTALLATION OF DOMESTIC WATER PIPING SYSTEMS.

3. PROVIDE SIZES AND TYPES MATCHING PIPING AND EQUIPMENT CONNECTIONS; PROVIDE FITTINGS OF MATERIALES WHICH MATH PIPE MATERIALS USED IN DOMESTIC WATER PIPING SYSTEMS.

4. PIPE SIZE 2" AND SMALLER TO BE COPPER TUBE. (A) WALL THICKNESS TYPE M, HARD-DRAWN TEMPER. AND (B) FITTINGS: WROUGHT COPPER, SOLDER JOINTS.

5. TUBE SIZE 2 - 1/2" AND LARGER TO BE COPPER TUBE; (A) WALL THICKNESS: TYPE L, HARD-DRAWN TEMPER AND (B) FITTINGS: WROUGHT-COPPER, SOLDER-JOINTS.

6. EXTERIOR WATER PIPING WITH PIPE SIZE 3" & OVER; (A) DUCTILE-IRON PIPE, WITH CEMENT-MORTAR LINING. (B) PIPING WEIGHT: SCHEDULE 150. (C) FITTINGS: DUCTILE-IRON, WITH RUBBER-GASKET JOINTS. (D) INSTALL IN ACCORDANCE WITH ANSI/AWWA C-60.

7. AT COMPLETION OF WATER SERVICE LINE INSTALLATION, FLUSH AND STERILIZE IN CONFORMANCE WITH AWWA C-601, TO SATISFACTION OF LOCAL AUTHORITIES HAVING JURISDICTION, AND PROVIDE A WRITTEN REPORT TO OWNER.

WASTE PIPING

1. SEWER PIPE ABOVE GROUND WITHIN STRUCTURE (A) CAST IRON SOIL PIPE WITH CAST IRON FITTING. (B) PVC HEAVY DUTY PIPING INLIEU OF CAST IRON. FOR UNDERGROUND AND STANDARD PVC ABOVE GRADE IF APPROVED BY THE CITY. (C) TYPE "M" COPPER PIPE AND TYPE "DWV" FITTINGS.

2. CAST IRON PIPE TO BE FIRST QUALITY, SMOOTH INSIDE AND OUT, OF STANDARD DIMENSIONS, FREE FROM CRACKS OR SAND HOLES, COATED WITH COAL TAR PATCH.

3. FITTINGS TO BE CAST IRON. STANDARD TYLER, CHARLOTTE PIPE, OR EQUAL

4. INSTALL PIPE AND FITTINGS TO REQUIRED GRADE WITH HUBS AND BOTTOM HALF SECTION IN DISTURBED SOIL.

5. GRADE ALL HORIZONTAL RUNS OF PIPE IN BUILDING AND UNDER FLOOR SLAB AT 1/4" PER FOOT DOWNWARD IN DIRECTION OF FLOW. IF IT IS ABSOLUTELY IMPOSSIBLE TO MAINTAIN A GRADE OF 1/4" PER FOOT, PIPING MAY SLOPE TO A MINIMUM GRADE OF NOT LESS THAN 1/8" PER FOOT WHERE IDENTIFIED ON DRAWINGS.

6. SUPPORT ALL HORIZONTAL RUNS OF CAST IRON SOIL PIPE IN BUILDING AT INTERVALS NO TO EXCEED 10'-0" AND AT EACH CHANGE OF DIRECTION. PROVIDE A SUPPORT AT THE BASE OF VERTICAL RISERS WITH INTERMEDIATE SUPPORTS AS REQUIRED.

7. PROVIDE CLEANOUTS IN AN ACESIBLE LOCATION AT BASE OF ALL RISERS IN SOIL, AND WASTE AND DRAIN PIPING AND AT EACH CHANGE IN DIRECTION IN HORIZONTAL RUNS OF PIPE. IN LONG STRAIGHT RUNS, PROVIDE A CLEANOUT AT LEAST EVERY 50 FEET.

8. PROVIDE A VENT FOR EACH TRAP OR AS SHOWN ON THE DRAWINGS. EXTEND EACH VENT VERTICALLY TO A POINT NOT LESS THAN 6" ABOVE THE EXTREME OVERFLOW LEVEL OF THE FIXTURE SERVED BEFORE OFFSETTING HORIZONTALLY MINIMUM OF 42" ABOVE FLOOR. WHENEVER TWO OR MORE VENT PIPES CONVERGE, EXTENT EACH SUCH PIPE AT LEAST 6" IN HEIGHT ABOVE THE TOP OF THE PLUMBING FIXTURE IT SERVICES BEFORE BEING CONNECTED TO ANY OTHER VENT.

9. EXTEND VENTS THROUGH THE ROOF A MINIMUM DISTANCE OF 12" AND TERMINATE AT LEAST 15' HORIZONTALLY FROM OPERABLE WINDOWS, DOORS, OR AIR INTAKES AND AT LEAST 3' ABOVE SUCH OPENINGS. FLASH/COUNTERFLASH ALL VENTS THROUGH ROOF AND COORDINATE WITH TYPE OF ROOF AND ARCHITECTURAL DETAILS AS SHOWN IN DRAWINGS.

10. TEST ALL PIPING IN ACCORDANCE WITH THE REQUIREMENTS OF LOCAL AUTHORITIES. TEST ALL SYSTEMS WITH WATER. TEST OF 10 FEET HEAD OF WATER. REPAIR LEAKS AND RETEST SYSTEM, REPEATING THIS PROCESS UNTIL PIPING SYSTEM IS FREE OF LEAKS.

GAS PIPING

1. COMPLY WITH APPLICABLE PROVISIONS OF ANSI B31.2 "FUEL GAS PIPING".

2. COMPLY WITH APPLICABLE PROVISIONS OF NFPA 54 (ANSI Z223.1) "NATIONAL FUEL GAS CODE."

3. COMPLY WITH REQUIREMENTS OF LOCAL GAS UTILITY COMPANY.

4. PROVIDE PIPING MATERIAL AND FACTORY-FABRICATED PIPING PRODUCTS OF SIZES, TYPES, PRESSURE RATINGS, AND CAPACITIES AS INDICATED. WHERE NOT INDICATED, PROVIDE PROPER SELECTION AS DETERMINED BY INSTALLER TO COMPLY WITH INSTALLATION REQUIREMENTS. PROVIDE MATERIAL AND PRODUCTS COMPLYING WITH ANSI B31.2 WHERE APPLICABLE, BASE PRESSURE RATING ON NATURAL GAS PIPING SYSTEM MAXIMUM DESIGN PRESSURES. PROVIDE SIZES AND TYPES MATCHING PIPING AND EQUIPMENT CONNECTIONS; PROVIDE FITTINGS OF MATERIALS WHICH MATCH PIPE MATERIALS USED IN NATURAL GAS PIPING SYSTEMS. WHERE MORE THAN 1 TYPE OF MATERIALS OR PRODUCTS ARE INDICATED, SELECTION IS INSTALLER'S OPTION. INSTALL PIPING AND PROVIDE SUPPORTS, ANCHORS, AND SEALS.

5. INSTALL GAS METER IN ACCORDANCE WITH LOCAL UTILITY COMPANY'S INSTALLATION INSTRUCTIONS AND COMPLY WITH REQUIREMENTS.

6. LABEL GAS PIPING YELLOW # IDENTIFY THE USE AS SPECIFIED ABOVE.

7. GAS SERVICE PIPING: (A) ALL PIPE SIZES; BLACK STEEL PIPE. (B) WEIGHT: SCHEDULE 40. (C) FITTINGS: WROUGHT-BUTT WELDING. (D) WRAPPING: MACHINE WRAP PIPE USING 50% OVERLAP WRAP, WITH POLYVINYL CHLORIDE TAPE. HANE WRAP FITTINGS USING 100% OVERLAP WRAP EXTENDING 6" BEYOND FITTING ONTO WRAPPED PIPE. COMPLY WITH TAPE MANUFACTURER'S INSTALLATION INSTRUCTIONS.

8. BUILDING DISTRIBUTION PIPING (EXPOSED PIPE ONLY); (A) PIPE SIZE 2" AND SMALLER: BLACK STEEL PIPE. (B) PIPE WEIGHT: SCHEDULE 40. (C) FITTINGS: MALLEABLE IRON THREADED.

9. PIPE SIZE 2-1/2" AND LARGER: BLACK STEEL PIPE (A) PIPE WEIGHT: SCHEDULE 40. (B) FITTINGS WROUGHT-BUTT WELDING.

10. SPECIAL VALVES REQUIRED FOR NATURAL GAS PIPING SYSTEMS INCLUDE THE FOLLOWING TYPES: (A) GAS COCKS 2" AND SMALLER: 150 PSI NON-SHOCK WOG, BRONZE STRAIGHTWAY COCK, FLAT OR SQUARE HEAD, THREADED ENDS. (B) GAS COCKS 2-1/2" AND LARGER: 125 PSI NON-SHOCK WOCK, IRON BODY BRONZE MOUNTED, STRAIGHTWAY COCK, SQUARE HEAD, FLANGED ENDS. (C) SUBJECT TO COMPLIANCE WITH REQUIREMENTS, MANUFACTURERS OFFERING GAS COCKS WHICH MAY BE INCORPORATED IN THE WORK INCLUDE THE FOLLOWING: DEZURIK, UNIT OF GENERAL SIGNAL, HOMESTEAD INDUSTRIES, INC.; VALVE DIV., JENKINS BROS., LUNKENHEIMER CO., DIV. OF CONVAL CORP., NIBCO, INC., POWELL CO., ROCKWELL MFG. CO., WALWORTH CO.

11. USE SEALANTS ON METAL GAS PIPING THREADS WHICH ARE CHEMICALLY RESISTANT TO NATURAL GAS. USE SEALANT SPARINGLY AND APPLY ONLY TO MALE THREADS OF METAL JOINTS.

12. REMOVE CUTTING AND THREADING BURRS BEFORE ASSEMBLING PIPING.

13. DO NOT INSTALL DEFECTIVE PIPING OR FITTINGS. DO NOT USE PIPE WITH THREADS WHICH ARE CHIPPED, STRIPPED OR DAMAGED.

14. PLUG EACH GAS OUTLET, INCLUDING VALVES, WITH THREADED PLUG OR CAP IMMEDIATELY AFTER INSTALLATION AND RETAIN UNTIL CONTINUING PIPING, OR EQUIPMENT CONNECTIONS ARE COMPLETED.

15. INSTALL DRIP LEGS IN GAS PIPING WHERE INDICATED AND WHERE REQUIRED BY CODE OR REGULATION.

16. INSTALL "TEE" FITTING WITH BOTTOM OUTLET PLUGGED OR CAPPED, AT BOTTOM OF PIPE RISERS.

17. USE DIELECTRIC UNIONS WHERE DISSIMILAR METAL ARE JOINED TOGETHER.

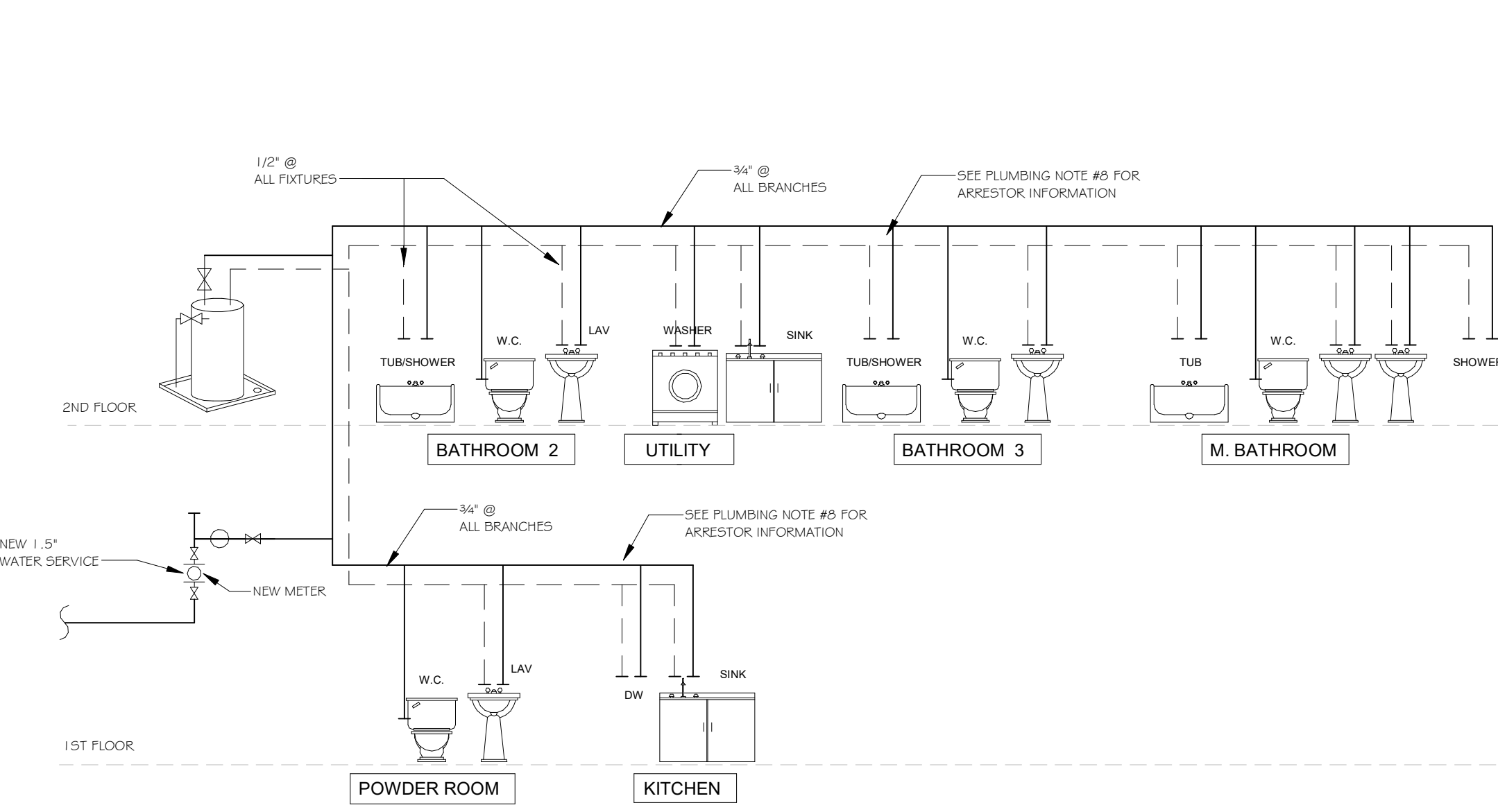
18. INSTALL PIPING WITH 1" DROP IN 60' PIPE RUN (0.14%) IN DIRECTION OF FLOW.

19. PROVIDE AT CONNECTION TO GAS TRAIN FOR EACH GAS-FIRED EQUIPMENT ITEM, AND ON RISERS AND BRANCHES WHERE INDICATED.

20. LOCATE GAS COCKS WHERE EASILY ACCESSIBLE AND WHERE THEY WILL BE PROTECTED FROM POSSIBLE INJURY.

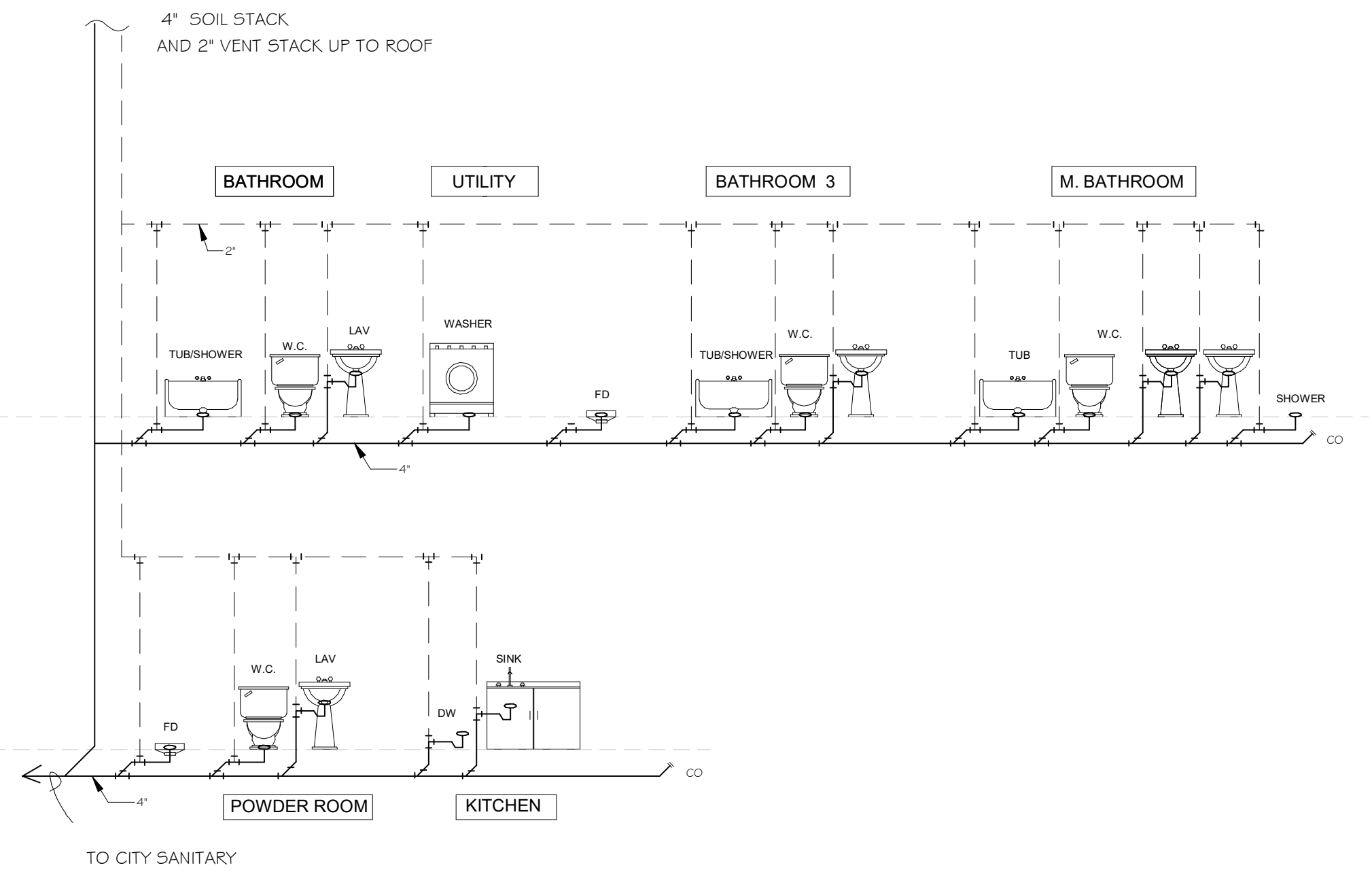
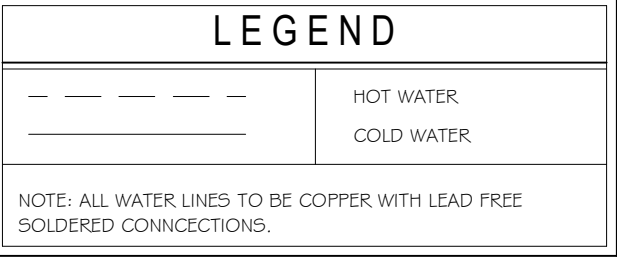
21. CONNECT GAS PIPING TO EACH GAS-FIRED EQUIPMENT ITEM, WITH DRIP LEG AND SHUTOFF GAS COCK. COMPLY WITH EQUIPMENT MANUFACTURER'S INSTRUCTIONS.

22. TEST NATURAL GAS PIPING IN ACCORDANCE WITH ANSI B31.2, AND LOCAL UTILITY REQUIREMENTS.

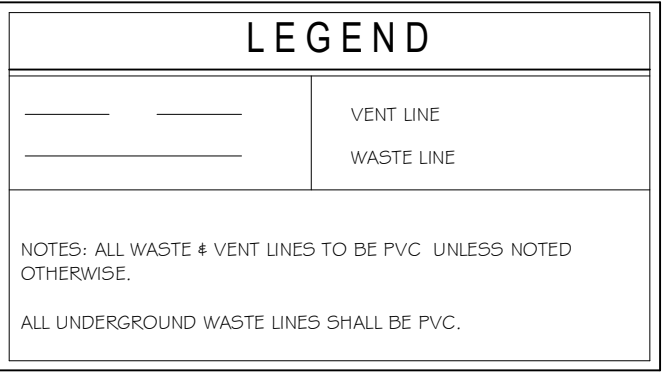


WATER SUPPLY DIAGRAM

NOTE: INSULATE ALL WATER LINES W/ 1/2" PIPE INSULATION.



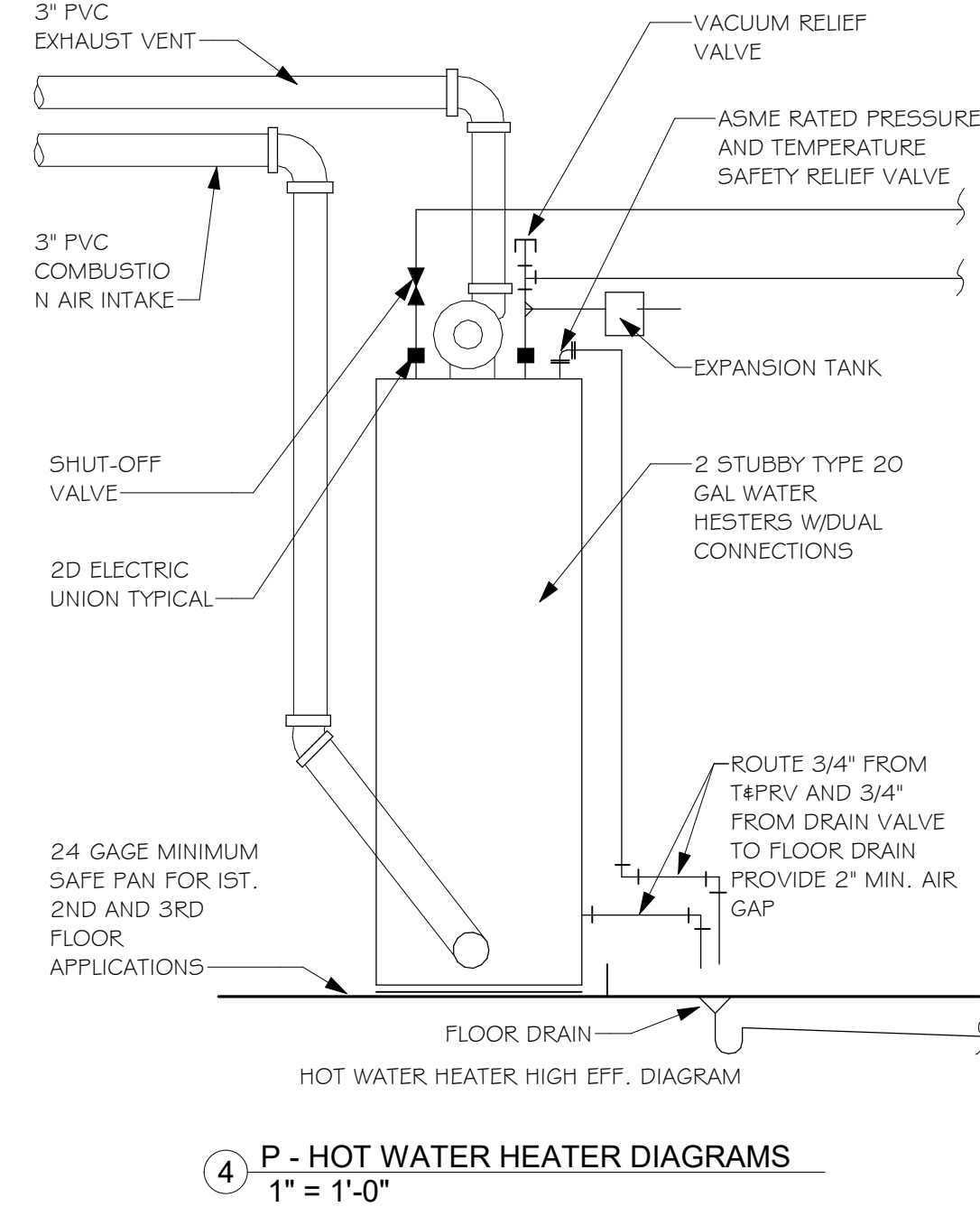
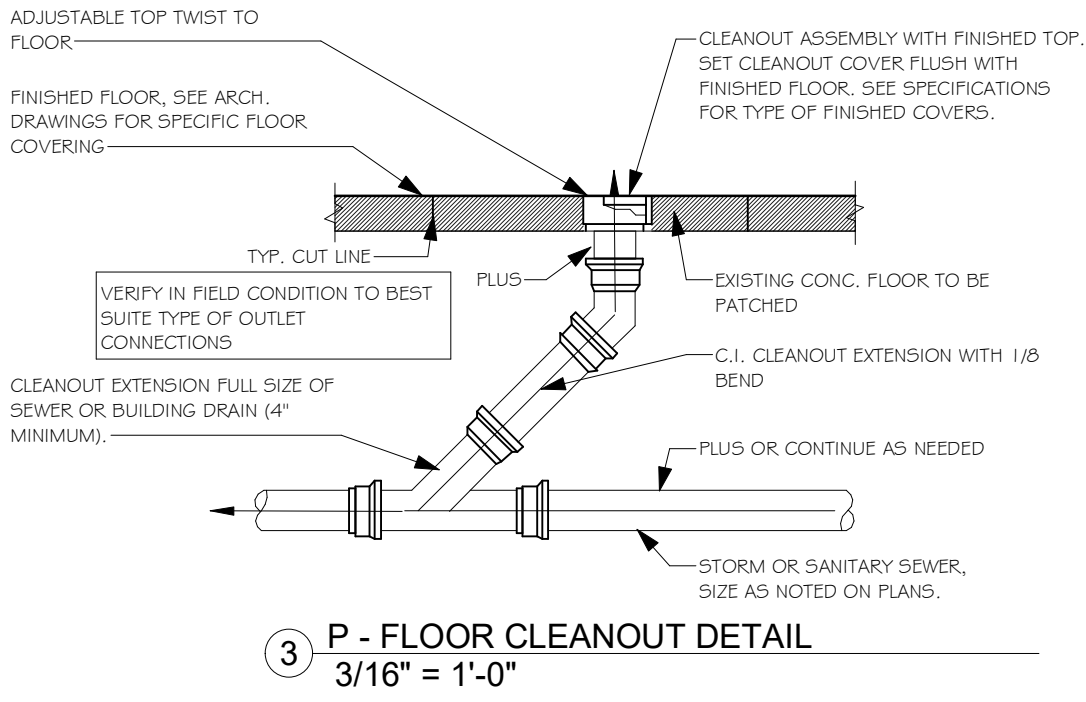
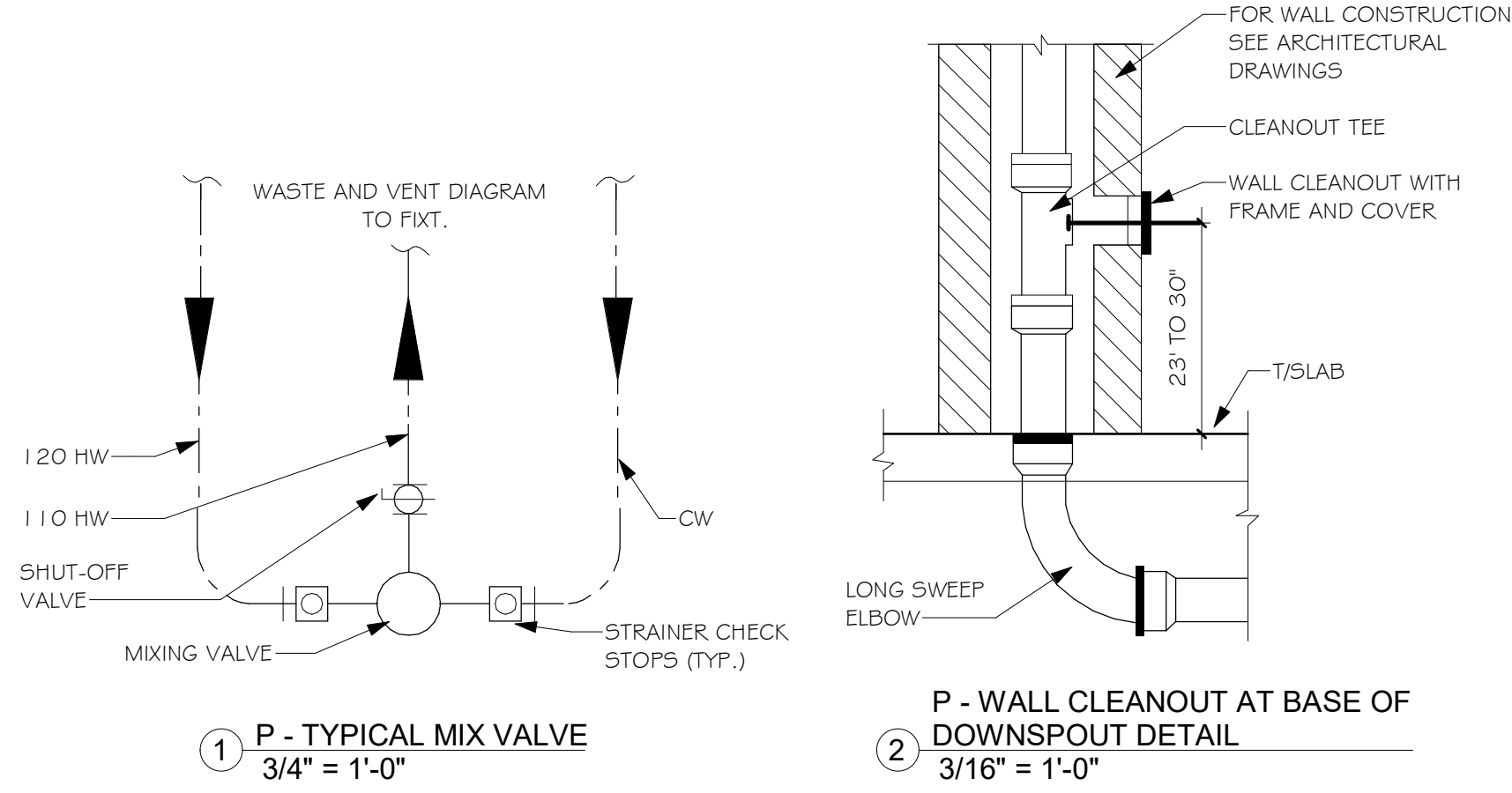
WASTE AND VENT DIAGRAM



Demand weight of Fixtures in Fixture Units.

Fixture Type	Occupancy Use	Valve Type	Fixture Units	WSFU	TOTAL	BASEMENT	1ST FLOOR	2ND FLOOR
						UNITS	UNITS	UNITS
Water closet	Private	Flush tank	5	X 3	15	1	2	2
Lavatory	Private	Faucet	6	X 1	6	1	2	3
Bathub/Shower	Private	Faucet/Mixing valv.	5	X 2	10	1	1	3
Kitchen sink	Private	Faucet	1	X 2	2	0	1	0
Dishwasher	Private	Faucet	1	X 2	2	0	1	0
Washing machine	Private	Faucet	1	X 2	2	0	0	1
Bidet	Private	Faucet	0	X 1	0	0	0	0

TOTAL FIXTURE UNITS	19	37 WSFU
TOTAL WATER SUPPLY FIXTURES UNITS		



ARCHITECTONIC SOLUTIONS, INC.

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CHICAGO, IL 60624
312-373-0098

DESIGN FIRM NO. 184.006860-0001

I HEREBY CERTIFY THAT THESE PLANS WERE PREPARED BY ME OR UNDER MY DIRECT SUPERVISION AND TO THE BEST OF MY KNOWLEDGE AND BELIEF, THEY COMPLY WITH ALL OF ROBBINS, ILLINOIS BUILDING CODES AND ORDINANCES



SIGNATURE: *Oscar Alva*

EXPIRES: 11/30/26

DATE	ISSUE
11/21/25	FOR REVIEW
1/12/26	FIELD REVISION

JOB TITLE

NEW
2 STORY FRAME
RESIDENCE

PROJECT ADDRESS

352 I 134th Pl.
Robbins, IL 60472

DRAWING TITLE

PLUMBING NOTES &
DIAGRAMS

PROJECT # 352 I

ISSUE DATE: 21/11/25

DRAWN BY: P.M.

CHECKED BY: O. ALBA

SHEET

P100